

Figure 8 shows a section of a suspension bridge. The bridge deck is supported by a single cable attached to a vertical tower.

Figure 8

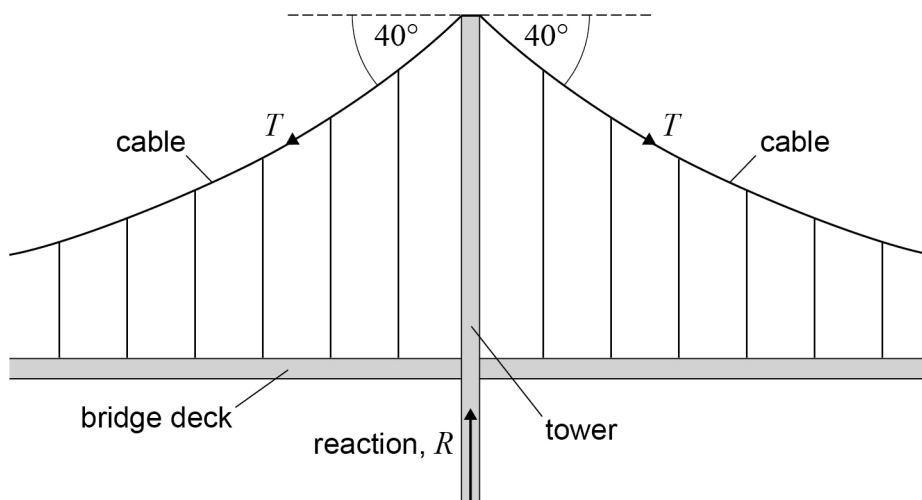
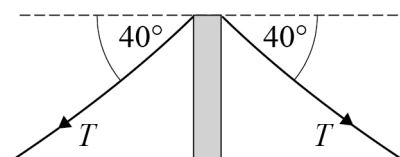


Figure 9 is an enlarged view of the cable and the top of the tower. The cable makes an angle of 40° to the horizontal where it meets the tower.

The tension T in each section of the cable is $1.2 \times 10^8 \text{ N}$
The weight of the cable is negligible.

Figure 9



08.1 Calculate the magnitude of the resultant force exerted on the tower by the cable.

[2 marks]

resultant force = _____ N



0 8 . 2 The mass of the tower is $7.1 \times 10^6 \text{ kg}$

Calculate the magnitude of the reaction R of the ground on the base of the tower.

[2 marks]

magnitude of R = _____ N

0 8 . 3 The cable has a cross-sectional area of 1.8 m^2 and an unstretched length of 1300 m

The cable is made from steel with a Young modulus of $2.1 \times 10^{11} \text{ Pa}$

Calculate the extension of the cable.

[3 marks]

extension = _____ m

Question 8 continues on the next page



0 8 . 4

The breaking stress of the steel cable is 4.7×10^8 Pa
However, it begins to undergo large plastic deformation at much smaller stresses.

Explain what is meant by breaking stress and by plastic deformation.

[2 marks]

breaking stress _____

plastic deformation _____

0 8 . 5

Explain why the bridge is designed so that, normally, the stress in the cable is significantly less than the breaking stress.

[1 mark]

Turn to page 14 for the next question



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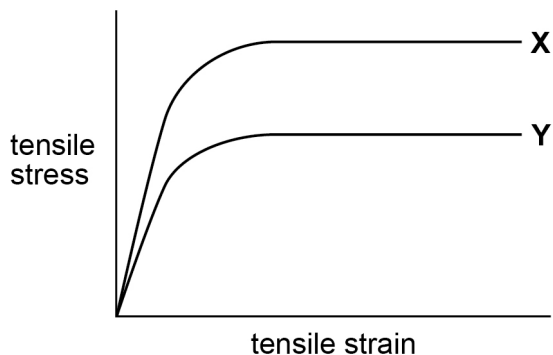


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0 4

Figure 1 shows the stress–strain relationship for two types of steel, **X** and **Y**.

Figure 1



0 4 . 1

State what is meant by tensile strain.

[1 mark]

0 4 . 2

Identify on **Figure 1** the limit of proportionality of steel **Y**. Label it **L**.

[1 mark]

0 4 . 3

An engineer plans to make a brake cable for a bicycle.

Explain why **X** is more suitable than **Y**.

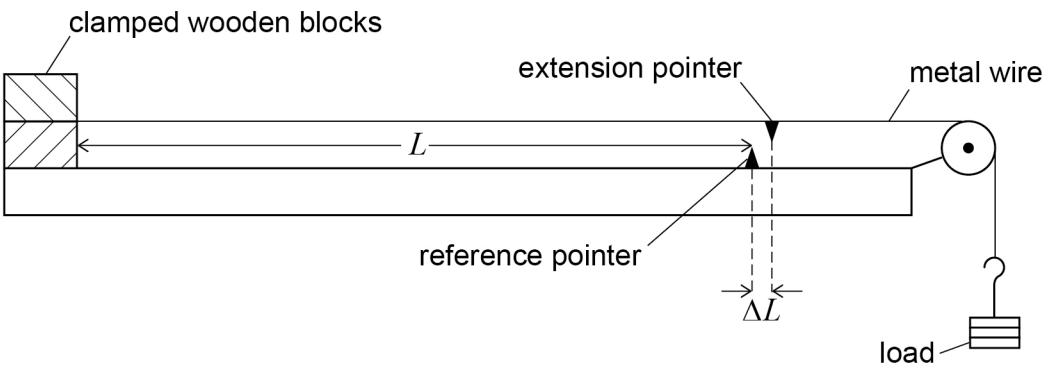
[3 marks]



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Figure 6 shows an apparatus used to measure the Young modulus of a metal.

Figure 6



The initial length L of a wire made from the metal was measured using a metre ruler. The mass of the load was increased in 50 g steps and the extension ΔL measured using vernier calipers. The experiment was carried out three times.

Table 2 shows some of the results of this experiment.

Complete Table 2.

[1 mark]

Table 2

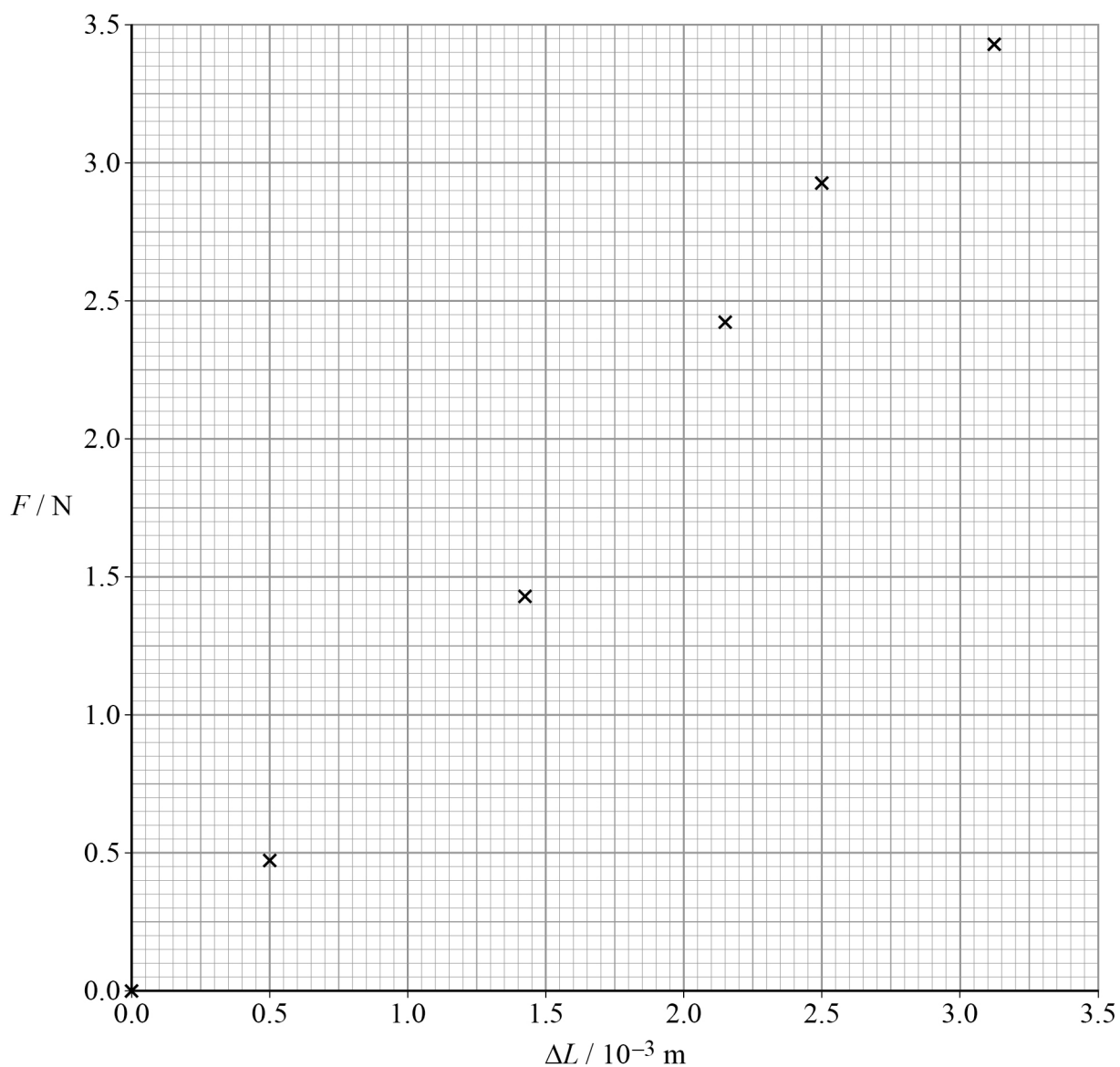
Tension F in the wire / N	$\Delta L / 10^{-3}$ m			
	First reading	Second reading	Third reading	Mean
0.00	0.0	0.0	0.0	0.0
0.49	0.5	0.5	0.5	0.5
0.98	0.8	0.7	0.9	
1.47	1.4	1.4	1.5	1.4
1.96	1.8	1.9	1.9	
2.45	2.2	2.1	2.3	2.2
2.94	2.5	2.5	2.5	2.5
3.43	3.1	3.1	3.2	3.1



- 1 0 . 2 Plot on **Figure 7** the two missing points.
Draw a best fit line.

[2 marks]

Figure 7



- 1 0 . 3 Determine the stiffness of the wire.

[2 marks]

stiffness = _____ N m^{-1}

Question 10 continues on the next page

Turn over ►



1 0 . 4

The cross-sectional area of the wire was $1.8 \times 10^{-8} \text{ m}^2$.
 L was 1.800 m.

Calculate the Young modulus of the metal.

[2 marks]

Young modulus = _____ Pa

7

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1**3**

Which is a unit for tensile stress?

[1 mark]

A $\text{kg m}^{-2} \text{s}^{-2}$

☐

B $\text{kg m}^{-1} \text{s}^{-2}$

☐

C kg m s^{-2}

☐

D $\text{kg m}^2 \text{s}^{-2}$

☐

Don't write

1

4

A student uses a micrometer to measure the diameter of a wire at several places along its length. These are the measurements he obtains:

0.58 mm

0.52 mm

0.50 mm

0.51 mm

0.56 mm

Which diameter should he quote for the mean value of the wire?

[1 mark]

A (0.5 ± 0.01) mm

☐

B (0.5 ± 0.1) mm

☐

C (0.53 ± 0.04) mm

☐

D (0.53 ± 0.08) mm

☐

2 3

Spring **P** of spring constant k is stretched until the tension is T and the energy stored is E .

Spring **Q** of spring constant $2k$ is stretched until the tension is also T .

What is the energy stored in **Q**?

[1 mark]

A $2E$

☐

B E

☐

C $\frac{E}{2}$

☐

D $\frac{E}{4}$

☐

Turn over ►



Don't write

01

A single nucleus of $^{11}_6\text{C}$ decays into a nucleus of $^{11}_5\text{B}$.
During this decay two additional particles are emitted.

01.1

Identify this decay.

[1 mark]

01.2

Identify the neutral particle emitted in this decay.

[1 mark]

01.3

Identify the antiparticle of the neutral particle in question 01.2.

[1 mark]

3



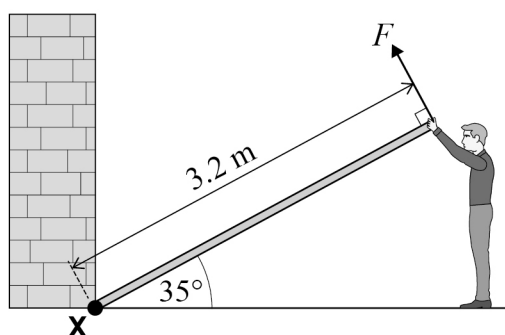
0 2 . 1 State the principle of moments.

[2 marks]

0 2 . 2

Figure 1 shows a uniform beam, of length 3.2 m and mass 27 kg. The beam is placed with one end against the bottom of a wall at **X**. A person holds the beam at 35° to the horizontal by applying a force F to the other end of the beam. F is applied at right angles to the beam.

Figure 1



Determine F by taking moments about **X**.

[3 marks]

$F =$ _____ N

0 3

A car has 280 kJ of kinetic energy when travelling at 80 km h^{-1} .

0 3

1

Calculate the mass of the car.

[3 marks]

mass = _____ kg

0 3

2

The car's brakes are applied. The car decelerates and comes to rest in a distance of 38 m.

Calculate the average resultant force acting on the car to bring it to rest.

[2 marks]

average resultant force = _____ N

5



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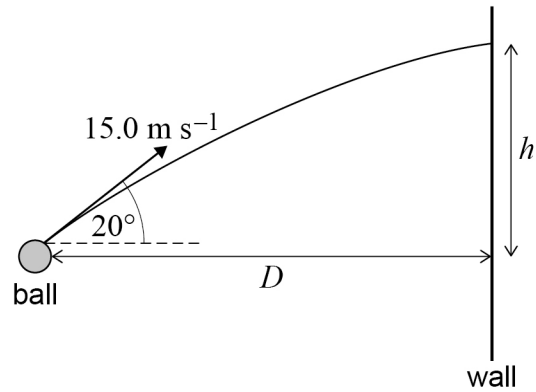
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0 4

Figure 2 shows a ball thrown with a velocity of 15.0 m s^{-1} at an angle of 20° to the horizontal. It reaches a wall 0.15 s after being thrown. Throughout this question, assume that air resistance is negligible.

Figure 2



0 4

1

Calculate the horizontal distance D travelled by the ball.

[2 marks]

$D =$ _____ m



0 4 . 2

Calculate the height h gained by the ball.

[3 marks]

$h =$ _____ m

0 4 . 3

Calculate the speed at which the ball hits the wall.

[3 marks]

speed = _____ m s^{-1}

8



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0 6

Figure 3 shows a sphere that falls through distance h onto horizontal ground. The sphere hits the ground and rebounds.

Figure 4 shows the variation with time t of the velocity v of the sphere.

Figure 3

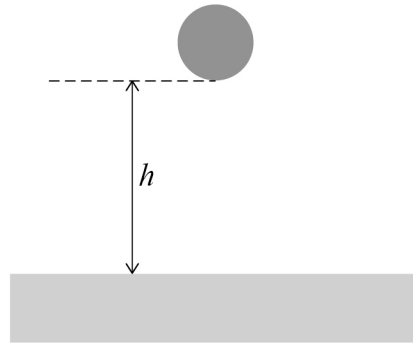
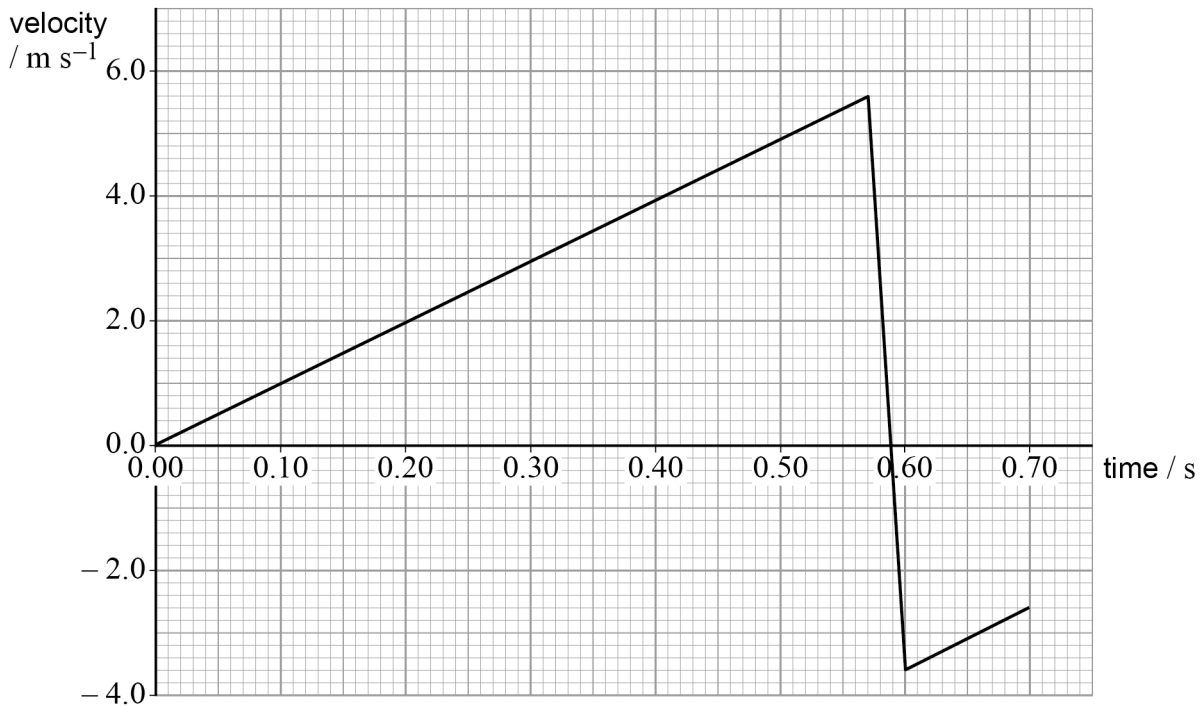


Figure 4



0 6

1

Calculate h .

[3 marks]

$h =$ _____ m



0 6 . 2

Calculate the magnitude of the acceleration of the sphere while it is in contact with the ground.

[3 marks]

acceleration = _____ m s^{-2}

0 6 . 3

The mass of the sphere is 58 g.

Calculate the force exerted by the ground on the sphere while it is in contact with the ground.

[3 marks]

force = _____ N

0 6 . 4

Another sphere of the same radius but much less mass is dropped in air from height h onto the ground. For this sphere, air resistance is significant during its descent.

Sketch onto **Figure 4** a velocity–time graph for this sphere from time $t = 0$ when it is dropped until it hits the ground for the first time.

[3 marks]

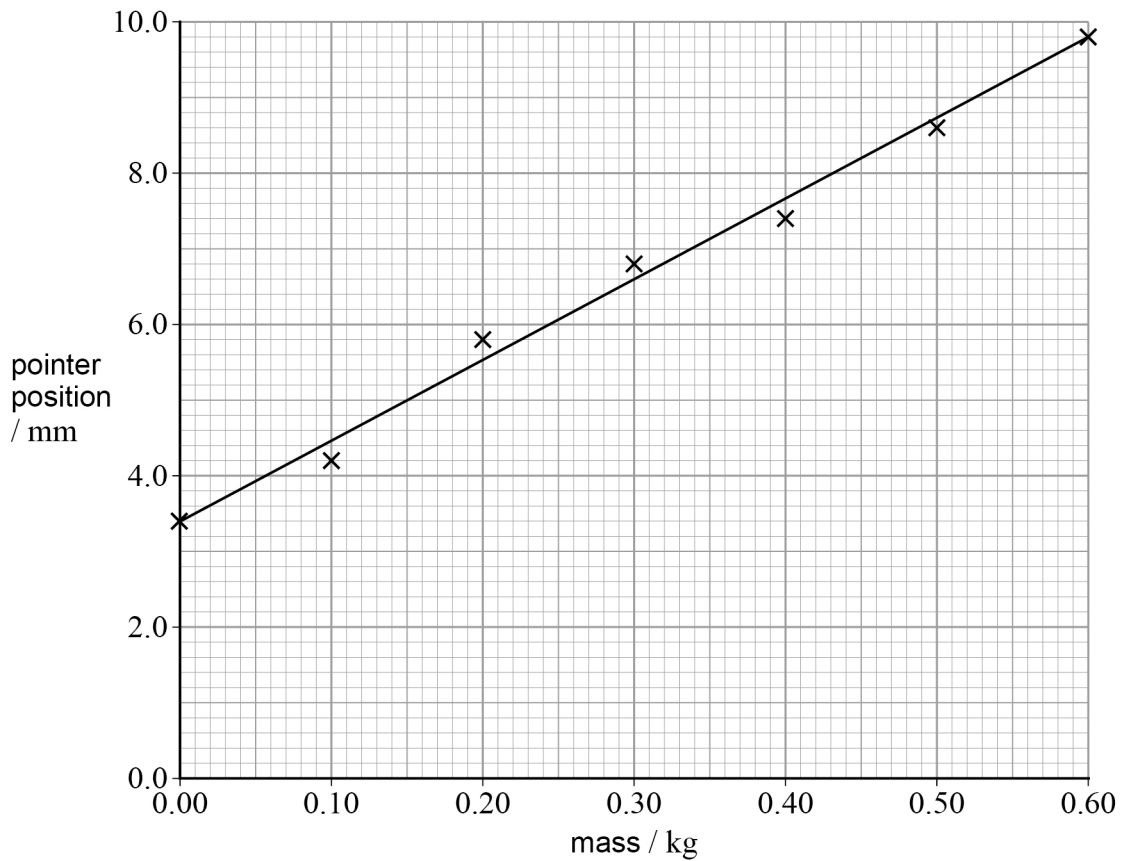


0 7

A wire hangs vertically from a fixed point with a mass attached to its bottom end. A student varies the mass hanging on the wire and measures the position of a pointer attached to the end of the wire.

Figure 5 shows the variation of pointer position with mass at the end of the wire.

Figure 5



0 7

1

Show that the stiffness of the wire is approximately 0.9 kN m^{-1} .

[4 marks]



0 7 . 2

The original unstretched length of the wire is 1.85 m and its cross-sectional area is $1.3 \times 10^{-5} \text{ m}^2$.

Calculate the Young modulus of the wire.
State an appropriate unit for your answer.

[3 marks]

Young modulus = _____

unit = _____

Question 7 continues on the next page



The wire is cut into two pieces, **X** and **Y**. The unstretched length of **X** is twice the unstretched length of **Y**. Equal masses are suspended from each wire.

0 7 . 3 Explain how the Young modulus of **X** compares with the Young modulus of **Y**.

[1 mark]

0 7 . 4 Explain how the stiffness of **X** compares with the stiffness of **Y**.

[2 marks]

0 7 . 5 Explain how the energy stored in **X** compares with the energy stored in **Y**.

[2 marks]

END OF SECTION A



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Section B

Answer **all** questions in this section.

0 8

Figure 6 shows two identical dummies **A** and **B** in a moving car. Dummy **A** has a seat belt. Dummy **B** does not have a seat belt.

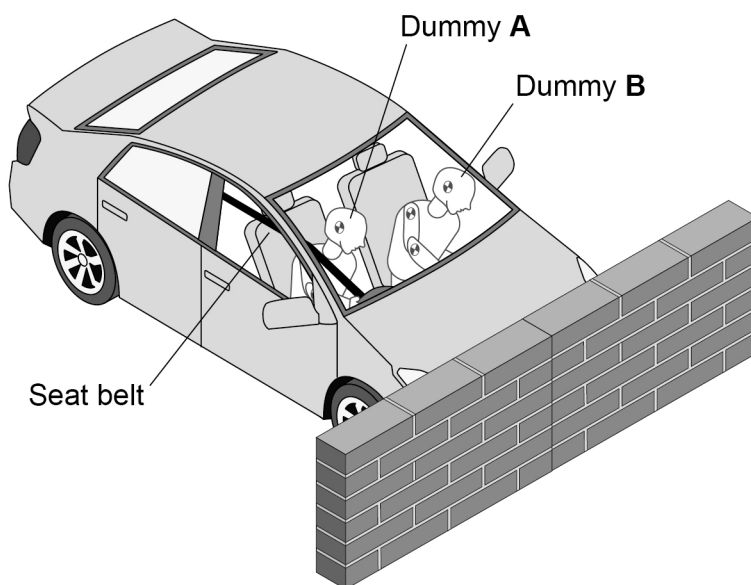
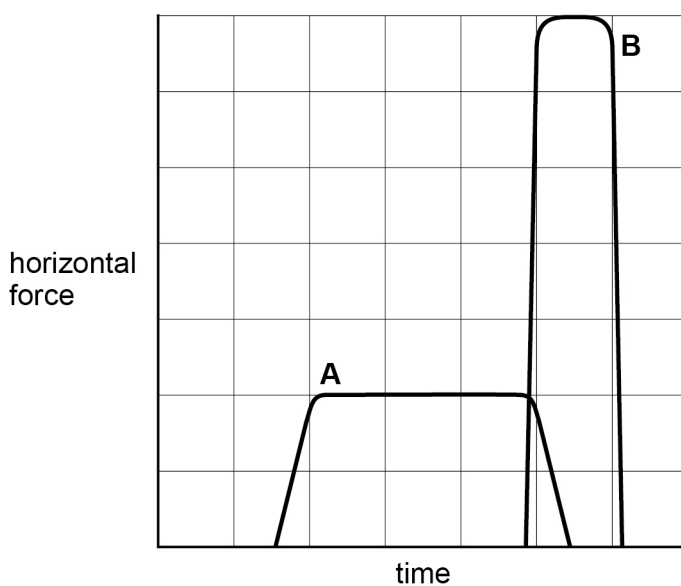
Figure 6

Figure 7 shows how the horizontal force on each dummy varies with time as the moving car is brought to rest.

Figure 7

0 8 . 1 The area under curve **A** is equal to the area under curve **B**.

Explain why.

[3 marks]

0 8 . 2 Explain why the same work is done on both dummies even though the maximum force on dummy **A** is less than the maximum force on dummy **B**, as shown in **Figure 7**.

[3 marks]

0 8 . 3 State Newton's first law of motion.

[1 mark]

0 8 . 4 Explain how Newton's first law of motion applies to dummy **B** which does not have a seat belt.

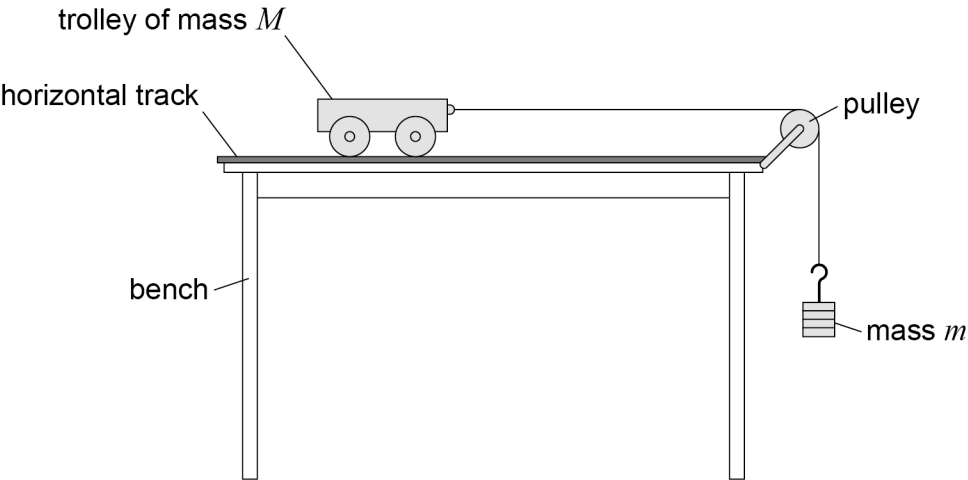
[1 mark]



0 9

Figure 8 shows the apparatus used in an experiment to illustrate Newton's second law of motion.
A trolley of mass M is accelerated by the weight of mass m .

Figure 8



The trolley and mass m are released from rest and travel a measured distance $s = 0.750$ m. The uncertainty in the measured distance $\Delta s = \pm 0.002$ m.

0 9 . 1

The time t taken for the trolley to travel distance s is measured five times.
Table 1 shows the results.

Table 1

t / s	0.81	0.77	0.80	0.79	0.78
----------------	------	------	------	------	------

Calculate the mean time t_{mean} and the uncertainty Δt_{mean} in these data.

[2 marks]

$t_{\text{mean}} = \text{_____ s}$

$\Delta t_{\text{mean}} = \text{_____ s}$



0 9 . 2 The acceleration a of the trolley is given by

$$a = \frac{2s}{t^2}$$

Calculate a and its percentage uncertainty.

[4 marks]

$$a = \text{_____} \text{ m s}^{-2}$$

percentage uncertainty in $a = \pm \text{_____}$

0 9 . 3 Newton's second law of motion when applied to the trolley and mass m gives

$$mg = (M + m) a$$

The mass of the connecting string is negligible.

Using this formula, a is predicted to be 2.56 m s^{-2} .

Explain how friction acting on the trolley can account for the difference between the predicted and the measured value of a .

[1 mark]

0 9 . 4 Suggest **one** improvement to the apparatus that could compensate for the friction acting on the trolley.

[1 mark]

END OF SECTION B



Section C

Each of the questions in this section is followed by four responses, **A**, **B**, **C** and **D**.

For each question select the best response.

Only **one** answer per question is allowed.

For each answer completely fill in the circle alongside the appropriate answer.

CORRECT METHOD



WRONG METHODS



If you want to change your answer you must cross out your original answer as shown. 

If you wish to return to an answer previously crossed out, ring the answer you now wish to select as shown. 

You may do your working in the blank space around each question but this will not be marked.
Do **not** use additional sheets for this working.

1 0 Which is a unit for stiffness?

[1 mark]

A J m^{-2}

☐

B J m^{-1}

☐

C J m

☐

D J m^2

☐

1 1 Which is usually a physical quantity with a unit?

[1 mark]

A accuracy

☐

B repeatability

☐

C reproducibility

☐

D resolution

☐

1 2 Which is a scalar quantity?

[1 mark]

A acceleration

☐

B charge

☐

C impulse

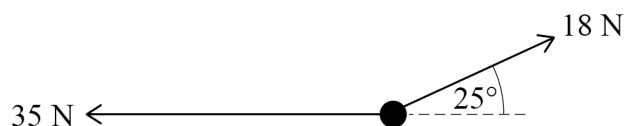
☐

D momentum

☐

1 3 What is the horizontal component of the resultant of these forces?

[1 mark]



A 51 N

☐

B 27 N

☐

C 19 N

☐

D 17 N

☐

Turn over for the next question

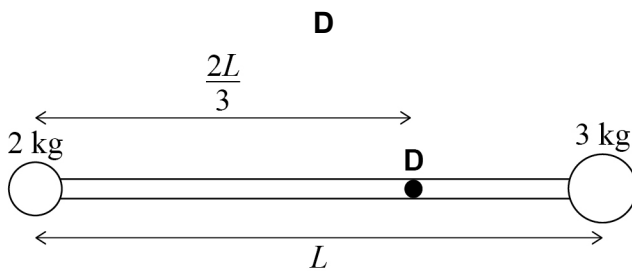
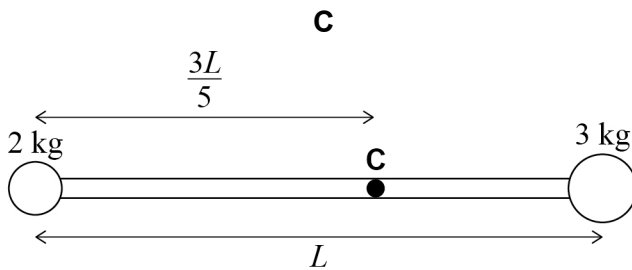
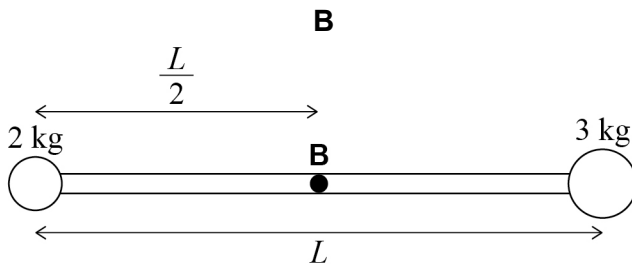
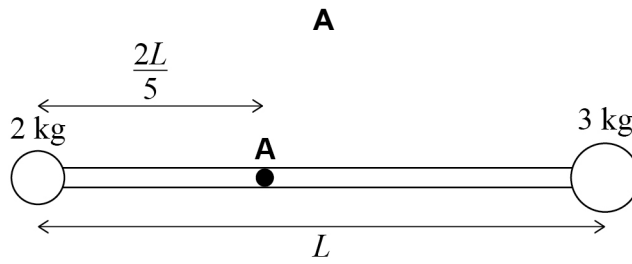
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1 4 A 2 kg mass and a 3 kg mass are joined by a rod of negligible mass and length L .

Which diagram shows the position of the centre of mass?

[1 mark]



A ☐

B ☐

C ☐

D ☐

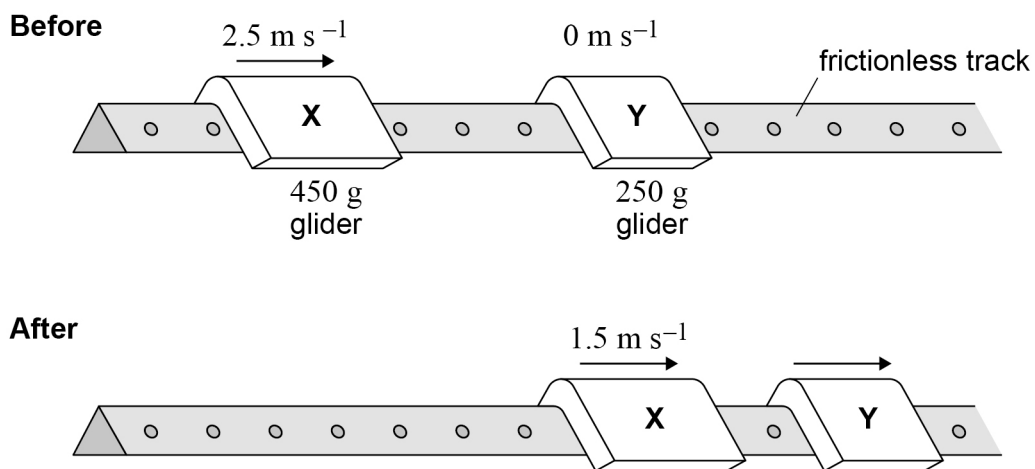


1 5

A glider **X** of mass 450 g is travelling at 2.5 m s^{-1} on a horizontal, frictionless air track. **X** collides with a stationary glider **Y** of mass 250 g. After the collision, **X** continues in the same direction with a speed of 1.5 m s^{-1} .

What is the speed of **Y** immediately after the collision?

[1 mark]



A 4.5 m s^{-1}

☐

B 1.8 m s^{-1}

☐

C 1.0 m s^{-1}

☐

D 0.9 m s^{-1}

☐**1 6**

What does the area under a force–displacement graph represent?

[1 mark]

A acceleration

☐

B energy transferred

☐

C impulse

☐

D spring constant

☐

1 7

A motor is 40% efficient and produces 96 W of useful output power.

What is the energy input to the motor in 2 minutes?

[1 mark]

A 240 J

☐

B 480 J

☐

C 4600 J

☐

D 29 000 J

☐

1 8

A car of mass m travels at a constant velocity v when its engine provides a useful output power P .

What is the resistive force acting on the car?

[1 mark]

A mv

☐

B $\frac{P}{v}$

☐

C Pmv

☐

D $\frac{Pv}{m}$

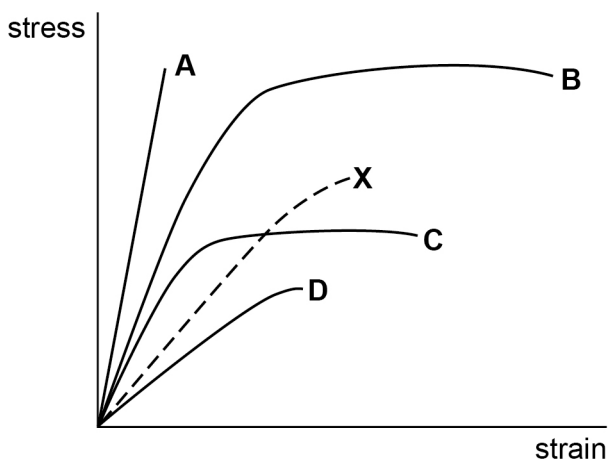
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1 9

Two metal specimens **X** and **Y** are stretched until they break. **Y** has a greater Young modulus, a greater ultimate tensile stress, and is less brittle than **X**. The stress–strain curve for **X** is shown.

Which is the stress–strain curve for **Y**?

[1 mark]



A ☐

B ☐

C ☐

D ☐

2 0

Solid spheres **A** and **B** have the same mass but different densities. The diameter of sphere **A** is $2d$ and the diameter of sphere **B** is d .

What is $\frac{\text{density of sphere A}}{\text{density of sphere B}}$?

[1 mark]

A 1 ☐

B $\frac{1}{2}$ ☐

C $\frac{1}{4}$ ☐

D $\frac{1}{8}$ ☐

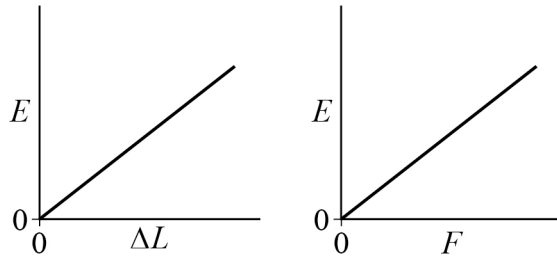
2 1

A spring that obeys Hooke's law is stretched.

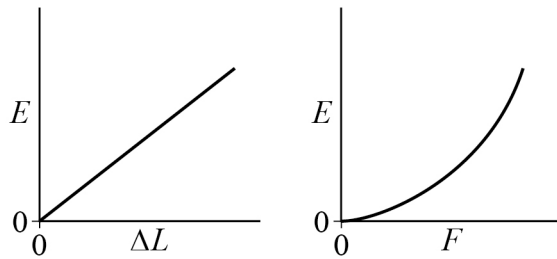
Which pair of graphs shows the variation of energy stored E with extension ΔL and the variation of E with force F ?

[1 mark]

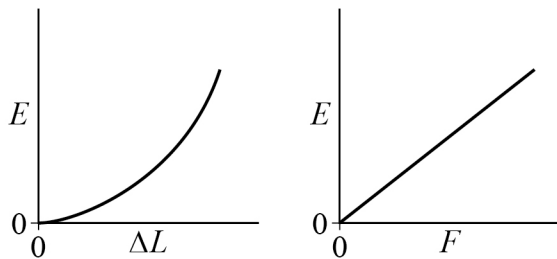
A



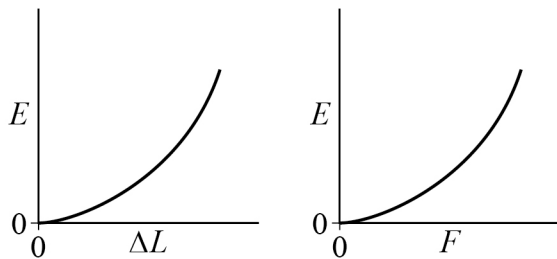
B



C



D



A ☐

B ☐

C ☐

D ☐

2 2

What is the specific charge of a ${}^9_4\text{Be}^{2+}$ ion?

[1 mark]

A $2.1 \times 10^7 \text{ C kg}^{-1}$ ☐

B $4.3 \times 10^7 \text{ C kg}^{-1}$ ☐

C $9.6 \times 10^7 \text{ C kg}^{-1}$ ☐

D $2.2 \times 10^{-1} \text{ C kg}^{-1}$ ☐

2 3

The initial activity of a radioactive source was 192 Bq. The activity of the source is 24 Bq after 12 days.

What was the activity of the radioactive source after 4 days?

[1 mark]

A 48 Bq ☐

B 80 Bq ☐

C 96 Bq ☐

D 108 Bq ☐

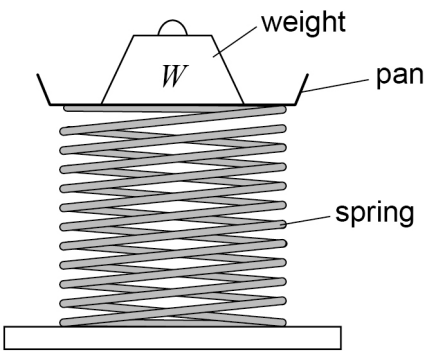
[illegible]

[illegible]

[illegible]

An engineer tests a spring for a car suspension system. The engineer measures the original length of the spring and then places a pan and weight W on top of the spring as shown in **Figure 8**.

Figure 8



The engineer measures the new length of the spring each time a weight is added. For each value of W the compression ΔL of the spring is calculated.

Table 2 shows the results of the experiment.

Table 2

W / N	50	100	150	200	250	300
$\Delta L / \text{mm}$	5.0	7.5	9.8	12.9	15.2	17.7

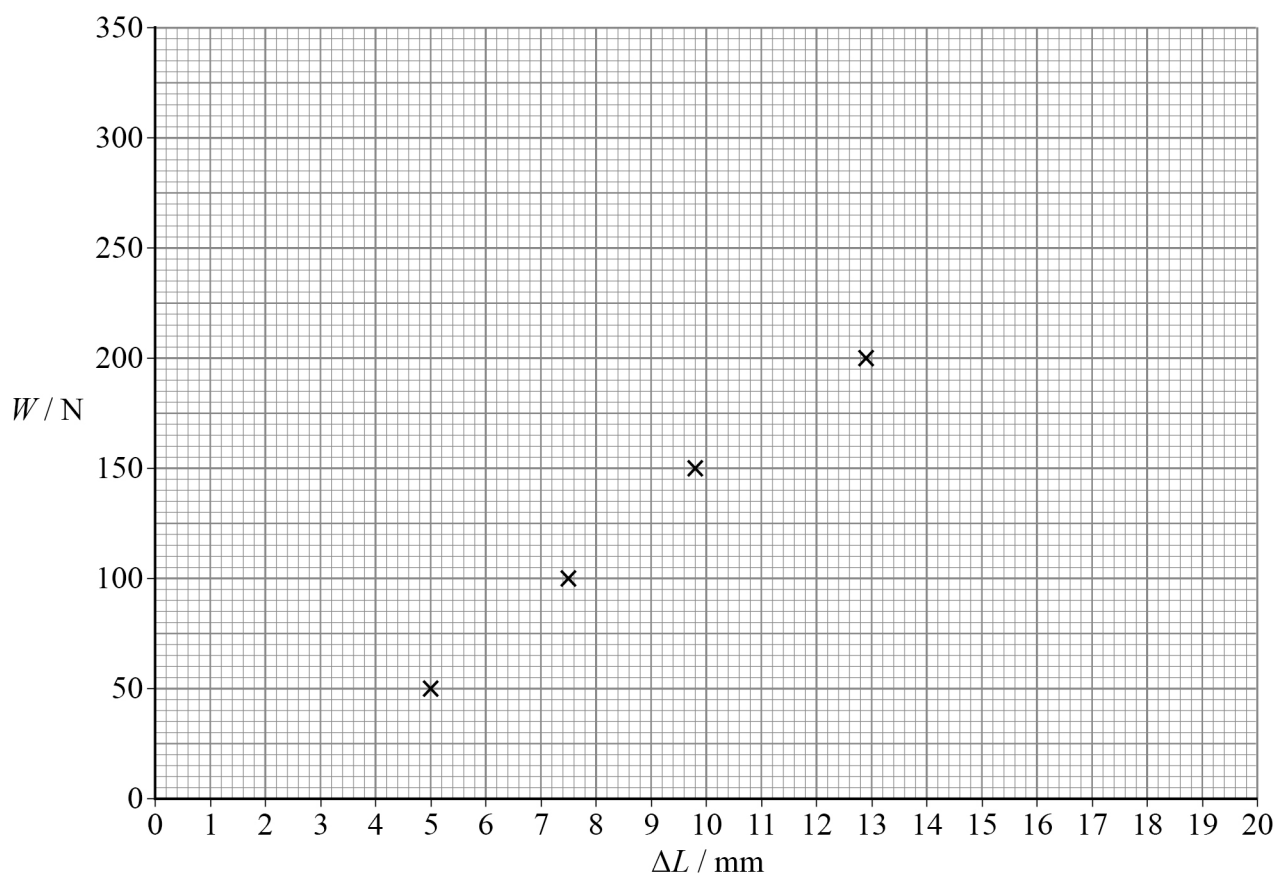
Some of these data points are plotted on the grid in **Figure 9**.

Plot the remaining **two** points and draw a line of best fit.

[2 marks]



Figure 9



1 1 . 2 Theory predicts that $W = k\Delta L$.

Identify the feature of the graph that shows that it does **not** support the theory.

[1 mark]

1 1 . 3 The graph does not support the theory because of a systematic error in the experiment.

State a possible systematic error that can account for this.

[1 mark]

Question 11 continues on the next page

Turn over ►



1 1 . 4

Determine the gradient of the graph.
State the unit for your answer.

[2 marks]

gradient = _____

unit = _____

1 1 . 5

Explain whether or not **Figure 9** can be used to find an accurate value for k in the equation $W = k\Delta L$.

[2 marks]



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A wire **W** has mass m , radius r and is made from a material of density ρ .
The masses, radii and densities for four other wires are shown in the table.

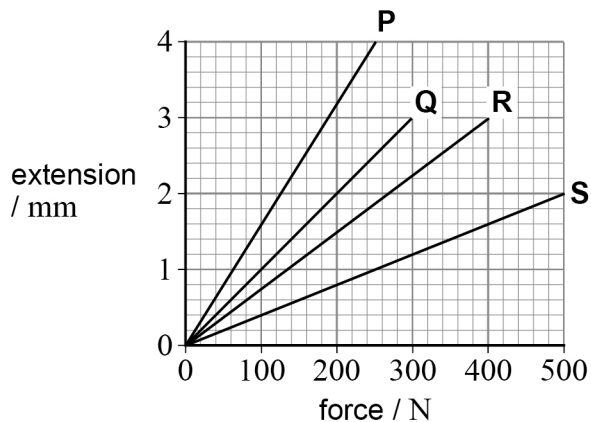
Which has the same length as **W**?

[1 mark]

	Mass	Radius	Density	
A	$2m$	$2r$	2ρ	☐
B	$4m$	$2r$	2ρ	☐
C	$4m$	$2r$	4ρ	☐
D	$8m$	$2r$	2ρ	☐

2**3**

The graph shows the variation of extension with force for four wires.



Which two wires store the same elastic strain energy at their maximum extension?

[1 mark]

A P and R

☐

B P and S

☐

C Q and R

☐

D Q and S

☐

Do not write

An electron is produced during pair production.

Describe what happens in the pair-production event.

[3 marks]



0 3 . 1

Alpha particles were directed towards a thin gold foil in the Rutherford scattering experiment.

Describe the paths of the alpha particles.

[3 marks]

0 3 . 2

The Rutherford scattering experiment was responsible for a change in the understanding of the structure of atoms.

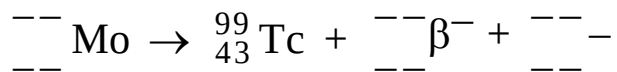
Describe how Rutherford's model of the atom differed from the previous model.

[2 marks]



0 4

A nucleus of molybdenum decays by β^- emission to produce technetium-99m as shown in the equation.



0 4 . 1

Complete the decay equation.

[3 marks]

0 4 . 2

Technetium-99m emits gamma radiation.

State the effect on the technetium-99m nucleus of the emission of a gamma ray.

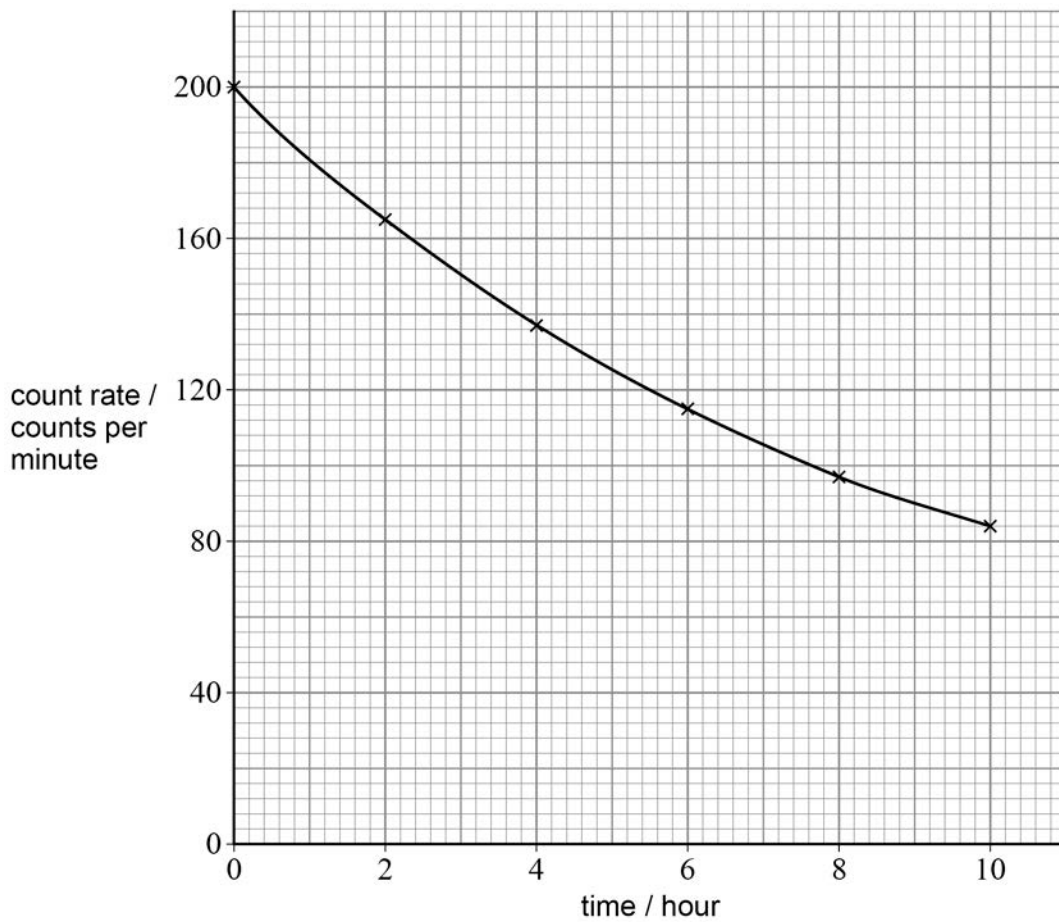
[1 mark]



A detector is placed near a source of technetium-99m.

Figure 1 shows the variation with time of the measured count rate.

Figure 1



The background count rate is 30 counts per minute.

Determine the half-life of technetium-99m.

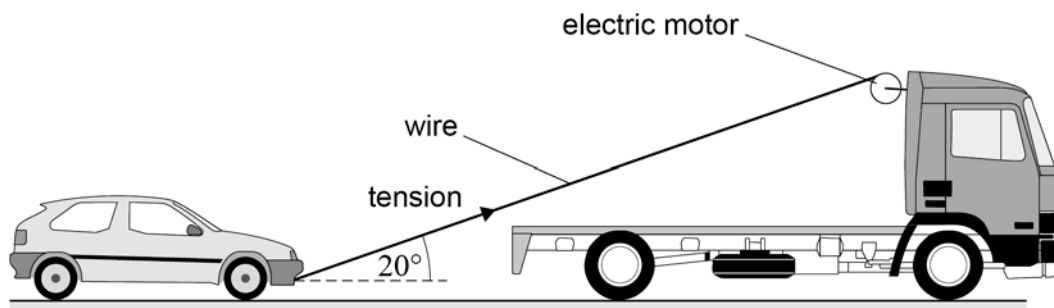
[3 marks]

half-life = _____ hour

0 5

A car is pulled along a horizontal road towards a stationary lorry. The car is pulled by a wire attached to an electric motor as shown in **Figure 2**.

Figure 2



At the instant shown in **Figure 2**, the power transferred by the electric motor to the car is 300 W.

The wire makes an angle of 20° to the horizontal.

The resistive forces acting on the car are constant throughout this question.

0 5

. 1

The car moves at a steady speed of 0.60 m s^{-1} .

Calculate the tension in the wire when the car is in the position shown in **Figure 2**.

[2 marks]

tension = _____ N



0 5 . 2 The motor has an efficiency of 87%.

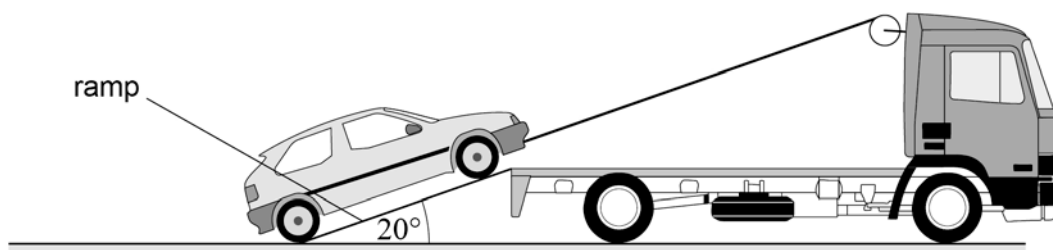
Calculate the electrical power input to the motor.

[2 marks]

electrical power input = _____ W

0 5 . 3 The car is then pulled up a ramp as shown in **Figure 3**.

Figure 3



The car is pulled up the ramp at a speed of 0.60 m s^{-1} .
The ramp and the wire both make an angle of 20° to the horizontal.

Explain why the power transferred by the electric motor to the car must be greater than 300 W.

[2 marks]

Question 5 continues on the next page

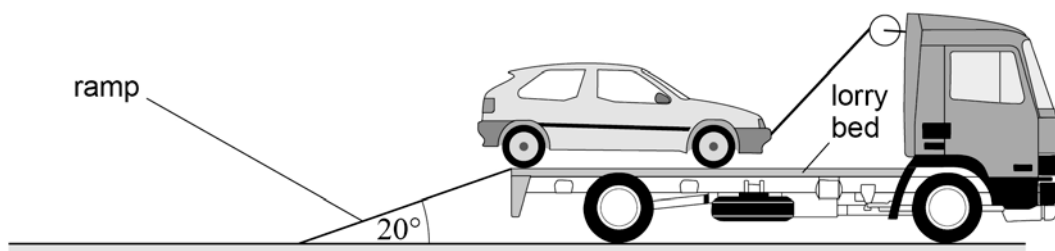
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0 5 . 4

The car is pulled along the horizontal lorry bed as shown in **Figure 4**. The tension in the wire remains constant as the car moves along the lorry bed.

Figure 4



State and explain what will happen to the speed of the car as it is pulled along the lorry bed.

[2 marks]



0 6 . 1

Explain the difference between speed and velocity.

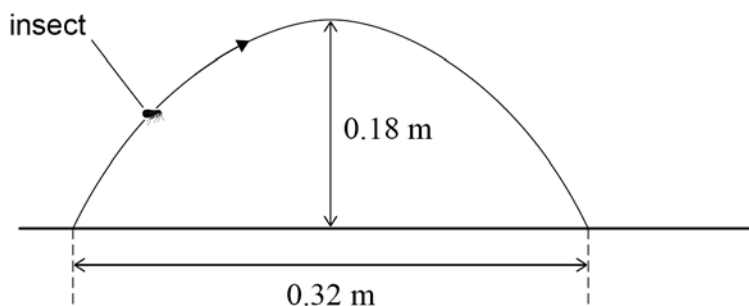
[2 marks]

Question 6 continues on the next page



When an insect jumps, it achieves a maximum height of 0.18 m and travels a horizontal distance of 0.32 m as shown in **Figure 5**. Air resistance is negligible.

Figure 5



0 6 . 2 Show that the initial vertical component of velocity of the insect is approximately 1.9 m s^{-1} .

[2 marks]

0 6 . 3 Show that the insect is in the air for approximately 0.4 s.

[3 marks]



0 6 . 4

State the position of the insect when it has its minimum speed during the jump.
Explain your answer.

[2 marks]

0 6 . 5

Show that the minimum speed of the insect during the jump is approximately 0.8 m s^{-1} .

[1 mark]

0 6 . 6

Determine the initial velocity of the insect.

[4 marks]

initial velocity = _____ m s^{-1}

direction _____



07

A climber **P** is connected to another person on the ground by a climbing rope as shown in **Figure 6**.

The rope runs over a frictionless pulley attached to the wall at **R**.

Figure 6



P falls and gravitational potential energy is transferred to work done in stretching the rope. The work done in stretching the rope is 5.3 kJ.

P is stationary for an instant at the end of the fall.

P has a mass of 80 kg.

07

. 1

Calculate the height through which **P** falls.

[2 marks]

height = _____ m



0 7 . 2 The climbing rope stretches by 1.6 m as **P** falls. The rope obeys Hooke's law.

Show that the stiffness k of the rope is approximately 4.0 kN m^{-1} .

[2 marks]

0 7 . 3 Calculate the tension in the rope when **P** is at the lowest position.

[1 mark]

tension = _____ N

Question 7 continues on the next page



07.4 After being brought to rest, **P** begins to accelerate upwards due to the tension in the rope.

Calculate the initial upward acceleration of **P**.

[3 marks]

acceleration = _____ m s^{-2}

07.5 The pulley at **R** is replaced with one that has a significant amount of friction.

Deduce how your answer to Question **07.4** would be different.

[3 marks]

END OF SECTION A



Turn over for Section B

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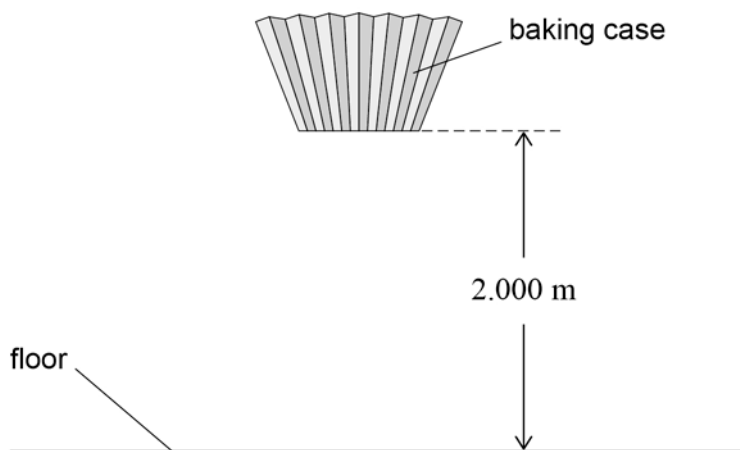
Section B

Answer **all** questions in this section.

0	8
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Figure 7 shows some of the apparatus used in an experiment to investigate how air resistance varies with speed. The air resistance is equal to the weight of a falling object when the object is at its terminal speed.

Figure 7



A small weight was placed in a baking case, giving a total weight w of 0.050 N . The baking case was dropped from a height of 2.000 m onto the floor. The time t for the baking case to fall this distance was measured.

Assume that the baking case fell at terminal speed v throughout the fall. The experiment was performed six times to find the mean time T for the fall.



08.1 Table 1 shows the results when w was 0.050 N.

Table 1

t/s	0.65	0.69	0.71	0.68	0.69	0.66
-------	------	------	------	------	------	------

The mean time $T = 0.68$ s.

State the absolute uncertainty in T .

[1 mark]

absolute uncertainty in $T =$ _____ s

08.2 Calculate v when $w = 0.050$ N.

[1 mark]

$v =$ _____ m s^{-1}

08.3 The uncertainty in the measurement of height is negligible.

Calculate the absolute uncertainty in your answer to Question **08.2**.

[2 marks]

absolute uncertainty in $v =$ _____ m s^{-1}

Question 8 continues on the next page

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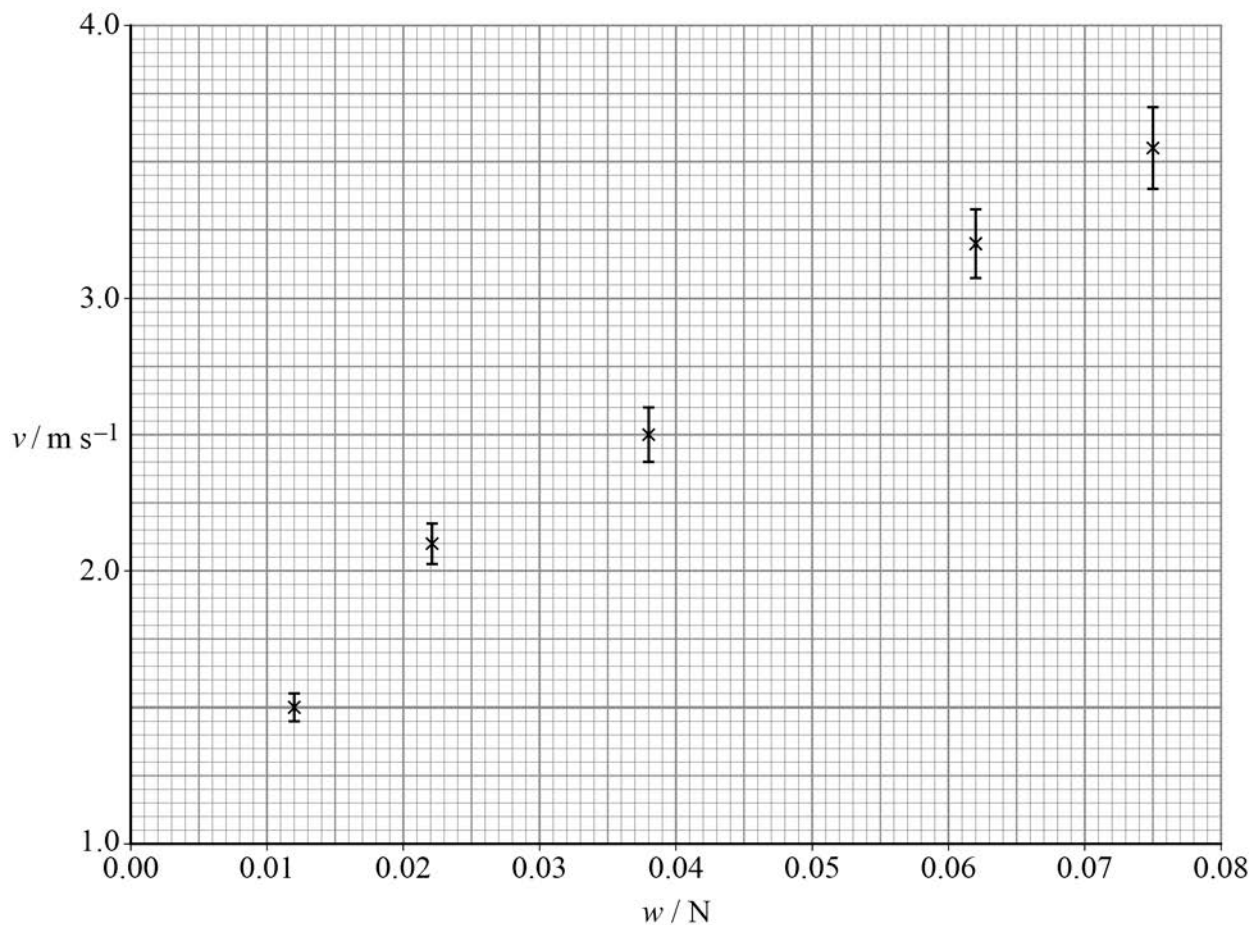
0 8 . 4

The experiment was then repeated with different weights placed in the baking case. For each weight the terminal speed v was calculated. The data are plotted on **Figure 8**.

Plot on **Figure 8** the data point for $w = 0.050 \text{ N}$ with its error bar.
Draw a line of best fit.

[2 marks]

Figure 8



0 8 . 5

It is suggested that, for a falling object, the force F acting on it due to air resistance is related to the instantaneous speed u of the object by

$$F \propto u^2$$

Explain how the results from this experiment could be used to test this suggestion.

[2 marks]

8

Turn over for the next question

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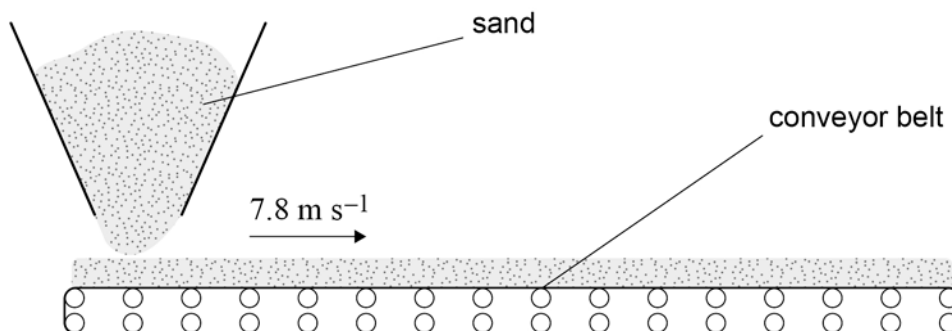


0 9 . 1 State what is meant by impulse.

[1 mark]

Sand falls vertically at a constant rate of $372 \text{ m}^3 \text{ min}^{-1}$ onto a horizontal conveyor belt as shown in **Figure 9**. The horizontal speed of the belt is 7.8 m s^{-1} .

Figure 9



0 9 . 2 The belt accelerates the sand to a speed of 7.8 m s^{-1} . The sand does not slip.

Show that the average horizontal force exerted by the belt on the sand is approximately $1.2 \times 10^5 \text{ N}$.

density of sand = 2500 kg m^{-3}

[3 marks]



0 9 . 3

The conveyor belt is driven by a motor continuously for a period of 24 hours.
Friction in the belt system is $4.5 \times 10^5 \text{ N}$.

Calculate the total energy transferred by the motor.

[3 marks]

total energy transferred = _____ J

0 9 . 4

As the sand lands on the belt, its horizontal component of momentum changes.

Explain how this change is consistent with the law of conservation of momentum.

[1 mark]

8

END OF SECTION B



Section C

Each of the questions in this section is followed by four responses, **A**, **B**, **C** and **D**.

For each question select the best response.

Only **one** answer per question is allowed.

For each question, completely fill in the circle alongside the appropriate answer.

CORRECT METHOD



WRONG METHODS



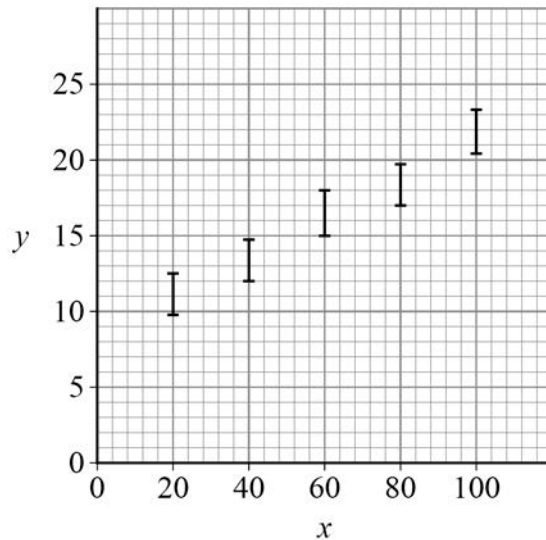
If you want to change your answer you must cross out your original answer as shown. 

If you wish to return to an answer previously crossed out, ring the answer you now wish to select as shown. 

You may do your working in the blank space around each question but this will not be marked.
Do **not** use additional sheets for this working.

1 0

The graph shows the variation of y with x .



What is the best value of the intercept on the y axis?

[1 mark]

A 7

☐

B 8

☐

C 10

☐

D 11

☐


1 1 A body in total equilibrium **must**

[1 mark]

A be travelling at a constant speed in a straight line.

☐

B be at rest.

☐

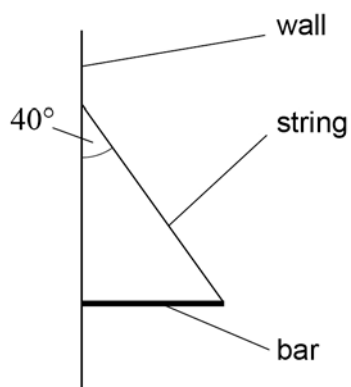
C have zero resultant force acting upon it.

☐

D have no couple acting upon it.

☐

1 2 A uniform bar of mass 1.3 kg is attached to a vertical wall and supported by a string. The string makes an angle of 40° to the vertical.



What is the tension in the string?

[1 mark]

A 8.3 N

☐

B 9.9 N

☐

C 16.6 N

☐

D 19.8 N

☐

1 3 An athlete runs at a constant speed of 5.0 m s^{-1} for 140 s around a circular track of circumference 200 m.

What is the magnitude of the athlete's final displacement from her starting point?

[1 mark]

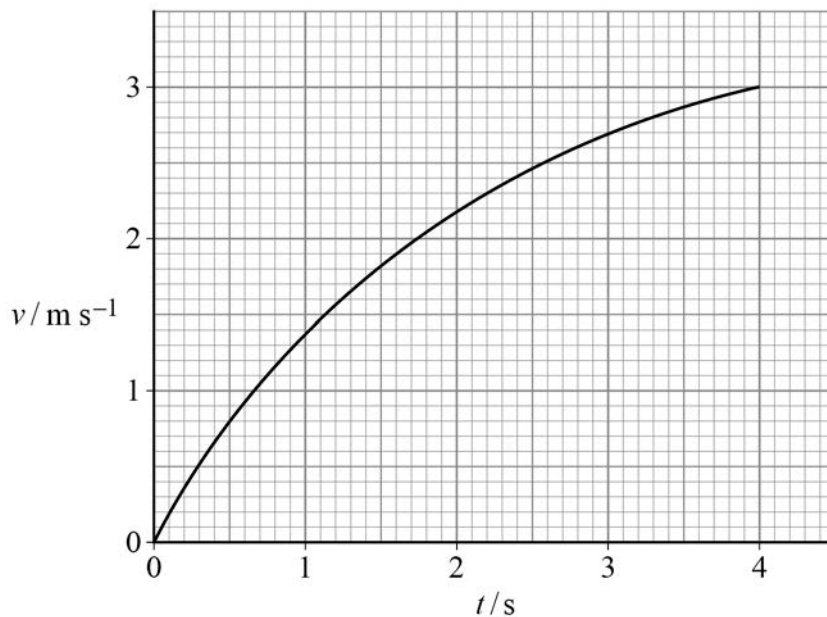
A 32 m ☐

B 45 m ☐

C 64 m ☐

D 100 m ☐

1 4 The graph shows the variation of speed v with time t for an object.



What are the initial acceleration and the total distance travelled for the object?

[1 mark]

	Initial acceleration / m s^{-2}	Total distance travelled / m	
A	0.75	6.0	☐
B	0.75	7.5	☐
C	2	6.0	☐
D	2	7.5	☐



1 5 Which statement is **always** correct?

[1 mark]

A Brittle fracture occurs with little or no elastic extension.

☐

B Brittle fracture occurs with little or no plastic extension.

☐

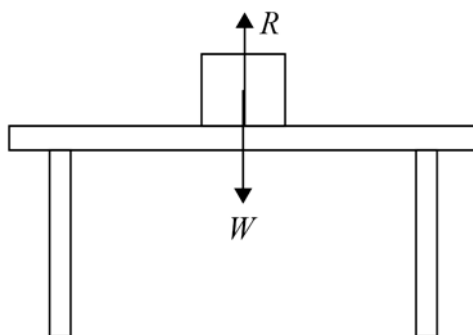
C In plastic behaviour, stress is proportional to strain.

☐

D In elastic behaviour, stress is proportional to strain.

☐

1 6 An object of weight W rests on a horizontal table. The normal reaction force of the table on the object is R .



Which statement is **not** true?

[1 mark]

A W and R are equal in magnitude.

☐

B The object is in equilibrium.

☐

C W and R are an action-reaction pair.

☐

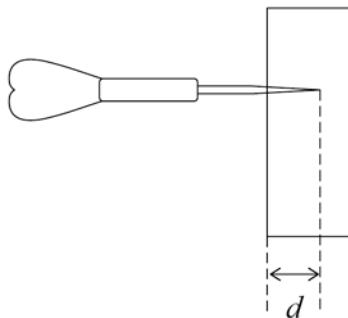
D R is a contact force.

☐


1 7

A dart of mass m hits a target with velocity v . The point of the dart penetrates a distance d into the target.

Assume that the deceleration of the dart is uniform.



The time taken for the dart to penetrate the distance d is

[1 mark]

A $\frac{d}{v}$ ☐

B $\frac{2d}{v}$ ☐

C $\frac{dm}{v}$ ☐

D $\frac{2dm}{v}$ ☐

1 8

Which list shows the particles in increasing order of their specific charges?

[1 mark]

A positron, ${}^4_2\text{He}$ nucleus, ${}^1_1\text{H}$ nucleus ☐

B neutron, ${}^4_2\text{He}$ nucleus, positron ☐

C ${}^1_1\text{H}$ nucleus, ${}^4_2\text{He}$ nucleus, neutron ☐

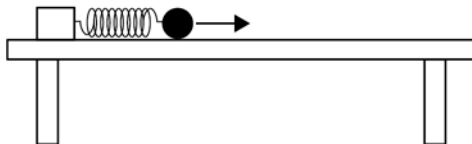
D proton, ${}^4_2\text{He}$ nucleus, neutron ☐



1 9 A steel ball of mass m is placed against a spring that is compressed by a distance ΔL . The ball is released so that it is projected along a smooth table top.

The spring constant is k .

The maximum speed of the steel ball is v .



The process is repeated with a spring of spring constant $3k$ and a second steel ball of mass $2m$. The spring is compressed by a distance ΔL .

The maximum speed of the second ball is

[1 mark]

A $\frac{\sqrt{2}}{3}v$

☐

B $\sqrt{\frac{2}{3}}v$

☐

C $\frac{2}{\sqrt{3}}v$

☐

D $\sqrt{\frac{3}{2}}v$

☐

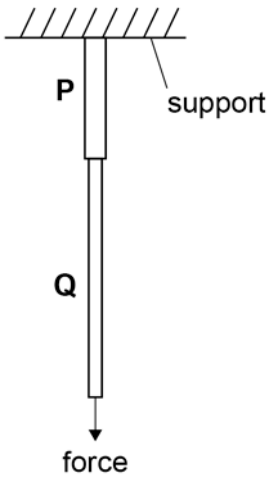
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2 0

A wire made of two sections **P** and **Q** is suspended from a support.
A force is applied to the end of the wire.



The table shows data for **P** and **Q**.

Property	P	Q
Young modulus	E	$3E$
Initial length	L	$2L$
Cross-sectional area	A	$\frac{A}{2}$
Extension	ΔL	

The **total** extension of the wire is

[1 mark]

A $\frac{2}{3}\Delta L$ ☐

B $\frac{4}{3}\Delta L$ ☐

C $\frac{5}{3}\Delta L$ ☐

D $\frac{7}{3}\Delta L$ ☐



2 1 $\bar{X}$ is the antiparticle of X .

Which properties are the same for X and $\bar{X}$?

[1 mark]

A mass and magnitude of charge only

☐

B mass and rest energy only

☐

C rest energy and magnitude of charge only

☐

D mass, magnitude of charge and rest energy

☐

2 2 A detector is placed 20 cm from a radioactive source in air. The corrected count rate is 50 counts per second.
A 4 mm aluminium absorber is placed between the source and the detector.
The corrected count rate then falls to 10 counts per second.

The emissions from the source could be

[1 mark]

A α , β^- and γ .

☐

B α and β^- only.

☐

C α and γ only.

☐

D β^- only.

☐

2 3 A radioactive isotope has a half-life of 8 years. The initial activity of the sample is 89 MBq.
After what time will the activity have fallen to 0.50 MBq?

[1 mark]

A between 40 years and 48 years

☐

B between 48 years and 56 years

☐

C between 56 years and 64 years

☐

D between 64 years and 72 years

☐

[illegible]

[illegible]

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07

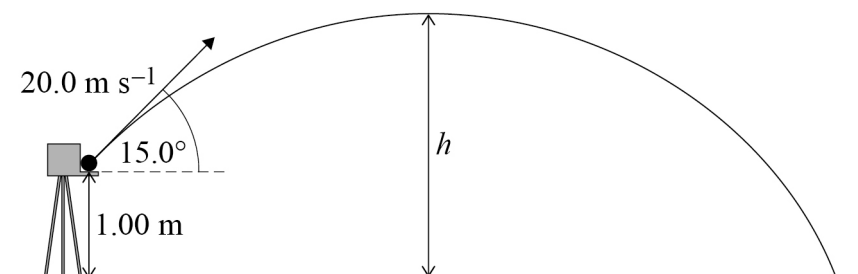
Figure 6 shows a machine that projects a tennis ball with an initial velocity of 20 m s^{-1} at an angle of 15.0° to the horizontal.

h is the maximum height that the ball reaches above the ground.

Ignore air resistance in all of your calculations.

Figure 6

not to scale



07

1

Calculate the vertical component of the initial velocity of the ball.

[1 mark]

vertical component = _____ m s^{-1}

07

2

Show that h is approximately 2.4 m.

[3 marks]



0 7 . 3

Calculate the horizontal distance the ball travels before it lands.

[3 marks]

distance = _____ m

0 7 . 4

The machine now projects a ball of mass 58 g vertically upwards.

The machine uses a compressed spring to do this.

The spring obeys Hooke's Law and is compressed through a distance of 0.050 m in order to project the ball.

The spring constant of the spring is 9.3 kN m^{-1} .

Calculate the maximum height above the point of release that the ball can reach.

[3 marks]

maximum height = _____ m

Question 7 continues on the next page

Turn over ►



0 7 . 5 The ball does not reach this maximum height.

Explain why.

[2 marks]

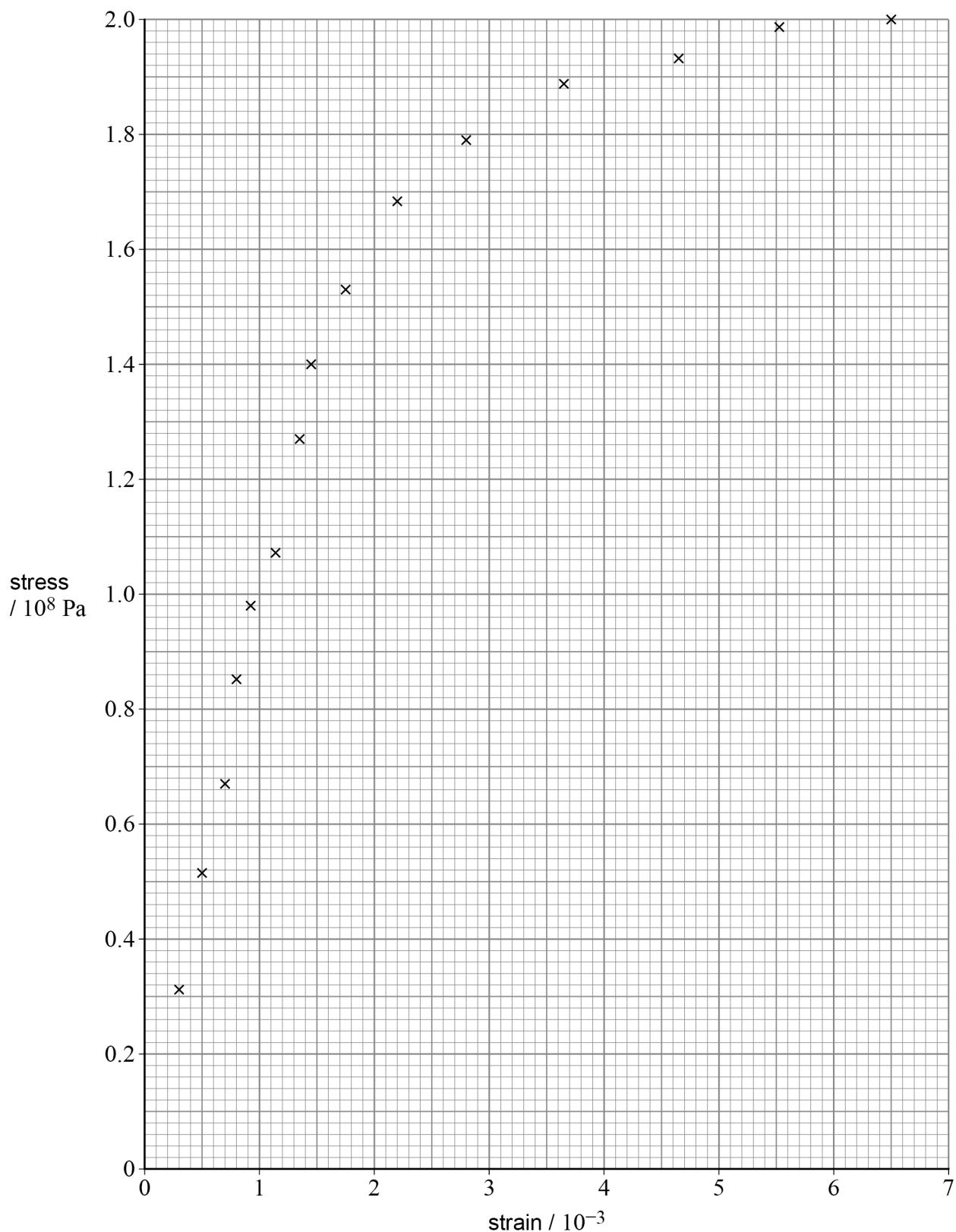


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Figure 12 shows the data from an experiment in which a copper wire was stretched by applying a known tensile force.

The total extension of the wire was recorded for each value of applied force.

Figure 12



1 0 . 1 Draw a line of best fit on **Figure 12**. **[1 mark]**

1 0 . 2 Determine, using **Figure 12**, a value for the Young modulus for copper. **[2 marks]**

Young modulus for copper = _____ Pa

1 0 . 3 Describe how the uncertainty in your value for the Young modulus could be estimated from the graph. **[1 mark]**

1 0 . 4 The area under a stress–strain graph represents the mechanical work done per unit volume on the material as it is deformed.
The original length of the wire used in this experiment is 1.50 m and its cross-sectional area is 0.397 mm².
Determine the work done on the wire as its original length is increased by 6.0 mm. **[4 marks]**

work done = _____ J

END OF SECTION B



Section C

Each of the questions in this section is followed by four responses, **A**, **B**, **C** and **D**.

For each question select the best response.

Only **one** answer per question is allowed.

For each question, completely fill in the circle alongside the appropriate answer.

CORRECT METHOD



WRONG METHODS



If you want to change your answer you must cross out your original answer as shown.

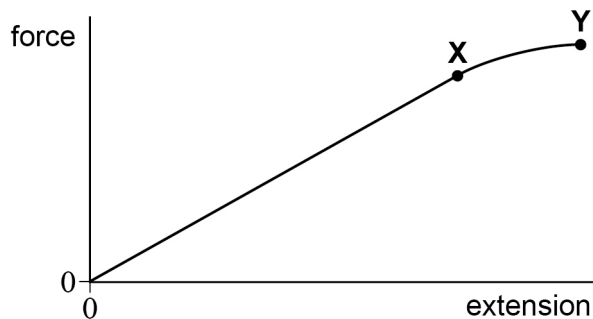


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You may do your working in the blank space around each question but this will not be marked.
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A force is applied to a metal wire and causes it to extend.
 The force is gradually increased from zero and then reduced back to zero.
 The force-extension curve is the same during loading and unloading.



What is the behaviour of the wire between 0 and X and between X and Y?

[1 mark]

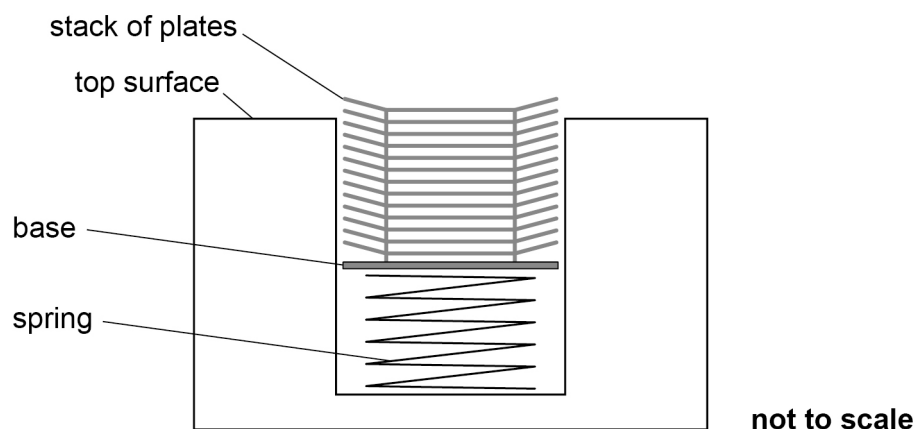
	0X	XY	
A	elastic	plastic	☐
B	elastic	elastic	☐
C	plastic	plastic	☐
D	plastic	elastic	☐

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06

Figure 3 shows a stack of plates in a plate holder used in a hotel kitchen. The stack of 13 plates is supported by a spring. The spring is compressed by 16.7 cm when there are 13 plates in the stack.

Figure 3



Each plate has a mass of 0.61 kg.

The spring obeys Hooke's law when supporting 13 plates.

The masses of the base and the spring are negligible.

06

.1

Calculate the stiffness of the spring.

[3 marks]

stiffness = _____ N m^{-1}



0 6 . 2 Calculate the energy stored in the spring when there are 13 plates in the holder.

[2 marks]

energy = _____ J

0 6 . 3 When a plate is added, the spring compresses so that the base moves downwards by a distance equivalent to the thickness of one plate. This ensures that the top plate is always just above the top surface of the holder.

Over time, the spring becomes weaker and its stiffness decreases.

Explain how the behaviour of the plate holder will be affected by the decrease in the stiffness of the spring.

[2 marks]

7

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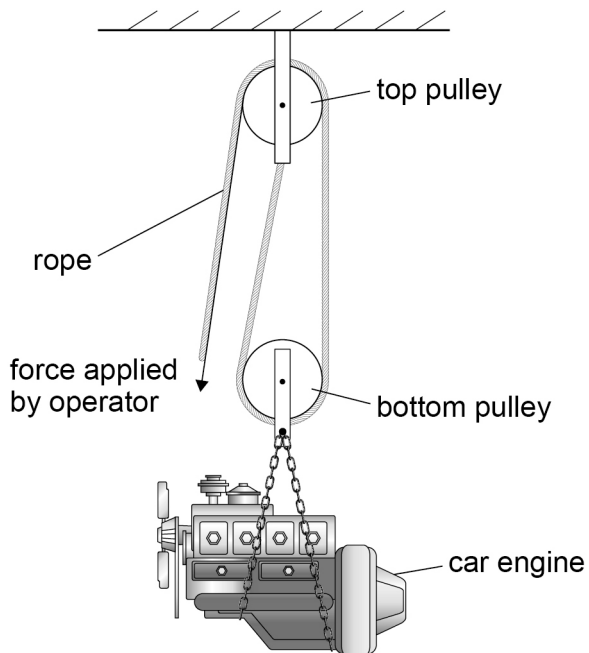
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Figure 6 shows a system of frictionless pulleys used to lift a car engine.

Figure 6



A single light rope is threaded around the top pulley and the bottom pulley. The operator applies a force to the free end of the rope. The bottom pulley and the car engine move upwards at a constant speed.

The engine has a mass of 140 kg.

The force applied by the operator is 720 N.

The engine moves vertically upwards by 1.2 m.

The extension of the rope is negligible compared with the distance moved by the engine.

Show that the work done by the operator is approximately 1700 J.

[2 marks]

0 9 . 2 Calculate the efficiency of the pulley system.

[2 marks]

efficiency = _____

0 9 . 3 Deduce the weight of the bottom pulley.

[2 marks]

weight = _____ N

0 9 . 4 The engine is now held in a fixed position.

The tension in the whole rope is 720 N.

The rope is made from a material with a Young modulus of 1.8×10^9 Pa.

The rope's cross-sectional area is $1.5 \times 10^{-4} \text{ m}^2$.

The unstretched length of the rope is 8.50 m.

Calculate the extension of the rope when the engine is held in the fixed position.

[2 marks]

extension = _____ m



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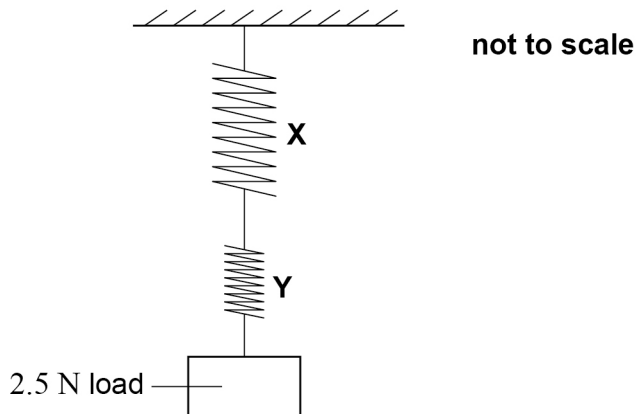
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In the diagram below, spring **X** has a spring constant of 15 N m^{-1} and an unstretched length of 8.0 cm .

Spring **Y** has a spring constant of 30 N m^{-1} and an unstretched length of 4.0 cm .

The load has a weight of 2.5 N .



The springs have negligible weight and obey Hooke's law.

What is the total length of **X** and **Y** when in equilibrium?

[1 mark]

A 0.18 m

☐

B 0.25 m

☐

C 0.30 m

☐

D 0.37 m

☐

07

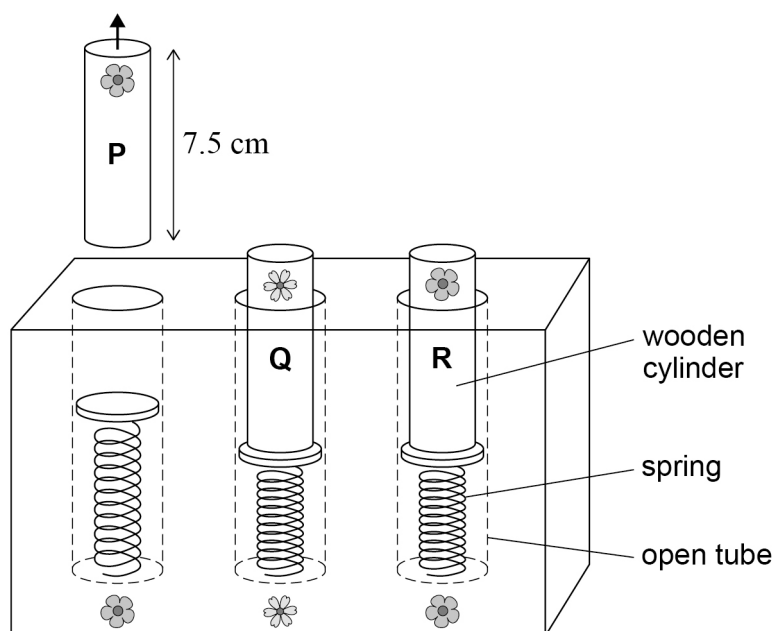
Figure 7 shows a toy with solid wooden cylinders **P**, **Q** and **R**. Identical springs are fixed at the bottom of the open vertical tubes.

The cylinders are placed into the tubes.

The weight of the cylinders causes negligible compression of the springs. The springs obey Hooke's law and air resistance is negligible.

A child pushes cylinder **P** down so that the spring is compressed. The child releases **P** and it is launched upwards.

Figure 7



07.1

P has length 7.5 cm, cross-sectional area 1.5 cm^2 and mass 8.1 g.

Calculate, in kg m^{-3} , the density of the wood.

[2 marks]

density = _____ kg m^{-3}



When a cylinder is launched, 60% of the elastic potential energy stored in the spring is transferred to the cylinder.

0 7 . 2

When the spring has been compressed by a distance x , the cylinder then rises through a height h .

Show that x and h are related by the following equation:

$$h = \frac{0.3 kx^2}{mg}$$

where k is the spring constant and m is the mass of the cylinder.

[2 marks]

0 7 . 3

Cylinders **P** and **Q** are identical.

The child pushes **P** down to compress the spring by 2.5 cm.

After release, **P** rises through a height of 17 cm.

The child then pushes **Q** down to compress the spring by 1.5 cm and releases it.

Calculate h for cylinder **Q**.

[2 marks]

$h =$ _____ cm

Question 7 continues on the next page

Turn over ►



0 7 . 4

Cylinder **R** is the same size as **P** but has half the mass.

The child launches **R**, using the same initial spring compression that was used for **P**.

Explain how h for **R** compares with h for **P**.

[2 marks]

8



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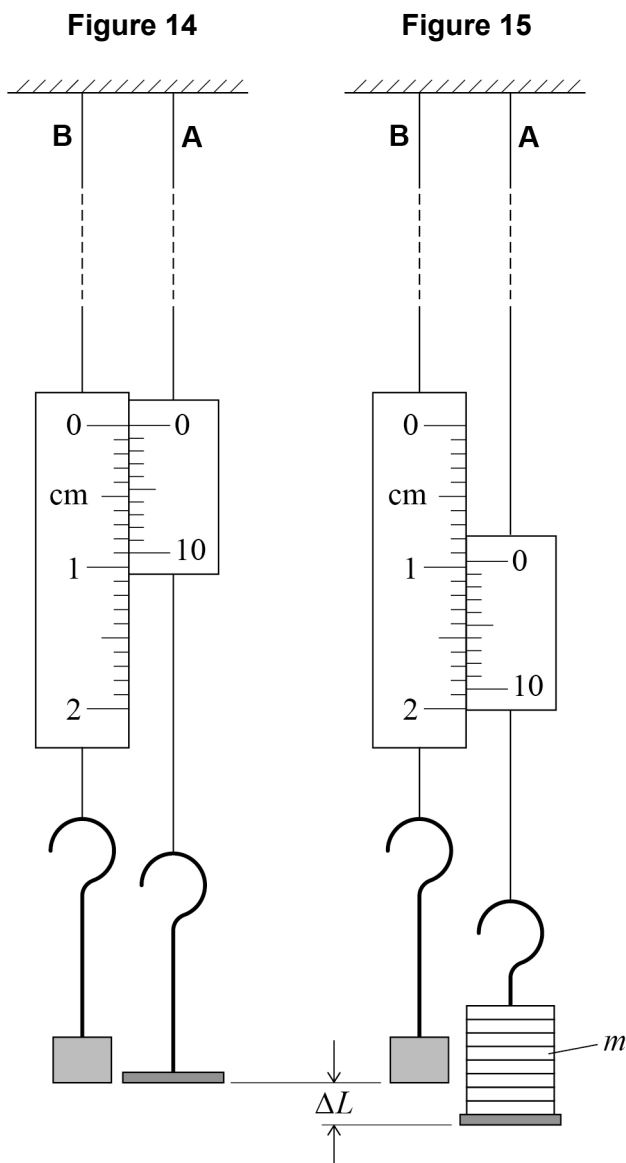
1 0

Figure 14 shows apparatus used to determine the Young modulus of a metal. Two identical wires **A** and **B** are made from this metal. **B** has a constant length. The length of **A** is compared with the length of **B**. The wires are suspended from a fixed support.

A student adds mass m to the load on **A** and measures the extension ΔL produced using a vernier scale. The student repeats this process for different values of m .

Figure 14 shows the vernier scale reading with $m = 0$

Figure 15 shows the vernier scale reading with $m = 4.0$ kg.



1 0 . 1

State the value of ΔL shown in **Figure 15**.

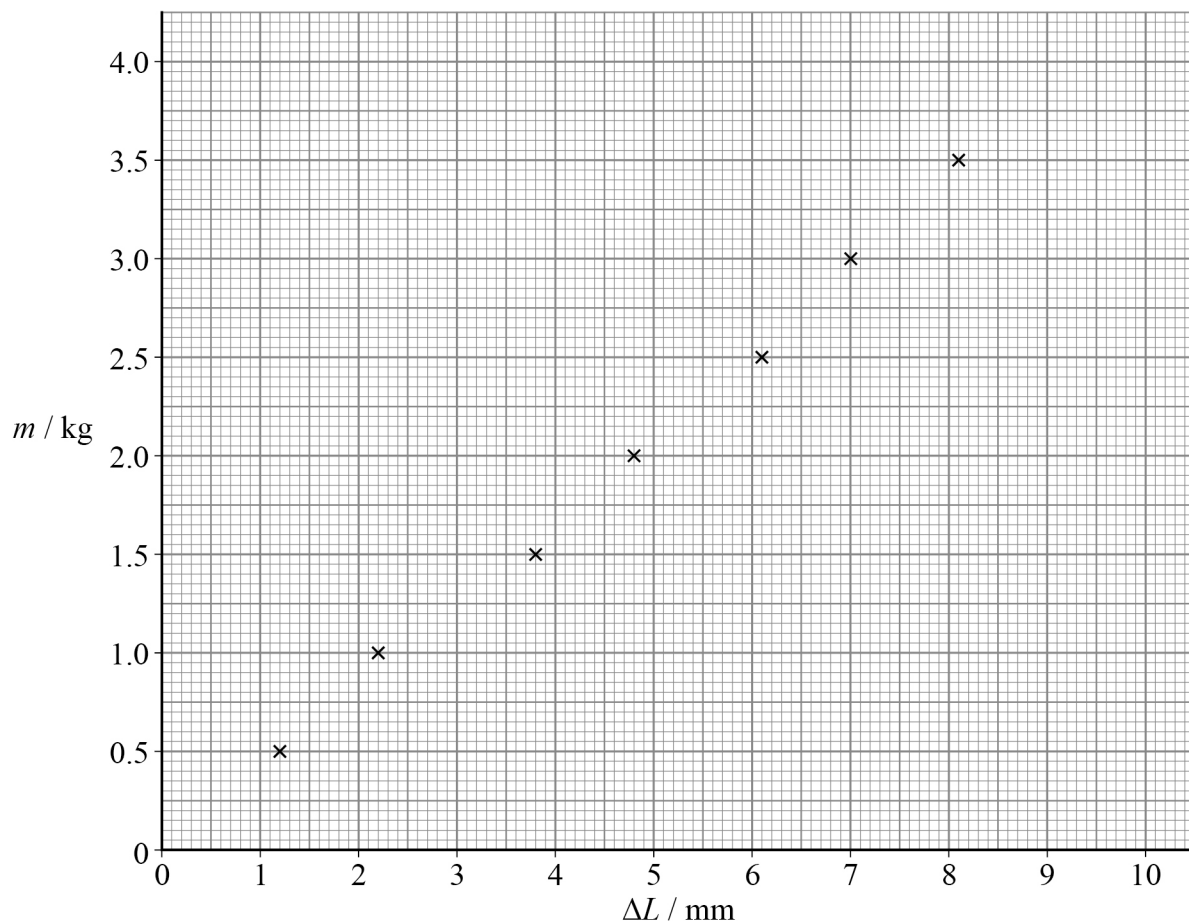
[1 mark]

$\Delta L =$ _____ mm



Some of the data collected by the student are plotted on the grid in **Figure 16**.

Figure 16



1 0 . 2 Plot, on **Figure 16**, your value of ΔL from Question **10.1**.

[1 mark]

1 0 . 3 Draw a line of best fit.
Determine, in kg m^{-1} , the gradient.

[3 marks]

gradient = _____ kg m^{-1}

Turn over ►



1 0 . 4

The original length of **A** is 2.600 m.
The cross-sectional area of **A** is $1.00 \times 10^{-7} \text{ m}^2$.

Determine the Young modulus of the metal.

[2 marks]

Young modulus = _____ Pa

1 0 . 5

Wire **A**, in the arrangement in **Figures 14** and **15**, is replaced with a thinner wire of the same metal.
The same measuring instruments are used.

Suggest how the uncertainty in the value of the Young modulus is affected by this change.

[2 marks]

END OF SECTION B



Section C

Each of the questions in this section is followed by four responses, **A**, **B**, **C** and **D**.

For each question select the best response.

Only **one** answer per question is allowed.

For each question, completely fill in the circle alongside the appropriate answer.

CORRECT METHOD



WRONG METHODS



If you want to change your answer you must cross out your original answer as shown.



If you wish to return to an answer previously crossed out, ring the answer you now wish to select as shown.



You may do your working in the blank space around each question but this will not be marked.
Do **not** use additional pages for this working.

1 7 Which is always a property of a brittle solid?

[1 mark]

Do not write
outside the
box

A It has a high stiffness.

☐

B It has a low breaking stress.

☐

C It stores little elastic strain energy before breaking.

☐

D It undergoes little plastic deformation before breaking.

☐

1**8**

Which quantity is given by the area under the stress–strain graph for a stretched material?

[1 mark]

A the energy stored

☐

B the energy stored per unit volume

☐

C the Young modulus

☐

D the Young modulus per unit volume

☐

Two wires **P** and **Q** are made of the same material.

The diameter of **P** is d and the diameter of **Q** is $2d$.

The length of **P** is $3l$ and the length of **Q** is l .

What is $\frac{\text{stiffness of P}}{\text{stiffness of Q}}$?

[1 mark]

A $\frac{1}{12}$



B $\frac{1}{6}$



C $\frac{3}{4}$



D $\frac{4}{3}$



Don't write

0	1
---	---

State the unit for tensile stress in SI fundamental (base) units.

[1 mark]

unit =

1

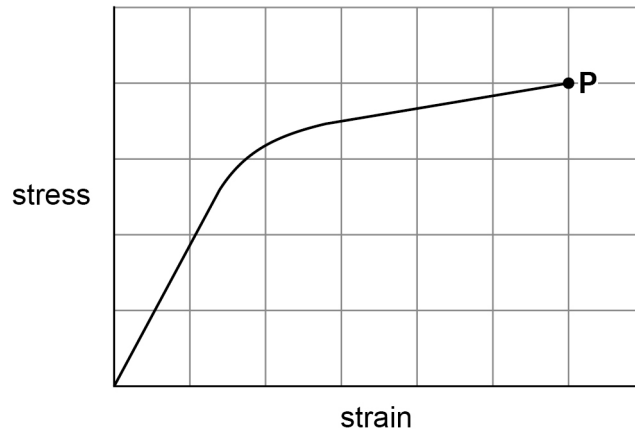
Figure 1 shows the variation of stress with strain for material **X**. Point **P** is the breaking point of material **X**.

Material **Y** is brittle. **Y** has a greater Young modulus than **X** and the same breaking stress as **X**.

Sketch, on **Figure 1**, the variation of stress with strain for **Y**.

[2 marks]

Figure 1



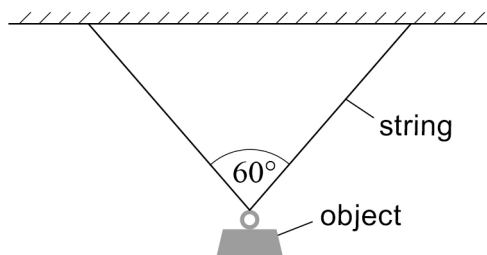
Turn over for the next question

Do not write

0 4

Figure 2 shows an object of mass 2.70 kg suspended by two identical light elastic strings. When the system is in equilibrium, the angle between the strings is 60° .

Figure 2



0 4 . 1

Calculate the tension in one string.

[2 marks]

tension = _____ N

0 4 . 2

The strain in each string is 1.5×10^{-2} .

The cross-sectional area of each string is 0.75 mm^2 .

Calculate the Young modulus of the material used to make the strings.

[2 marks]

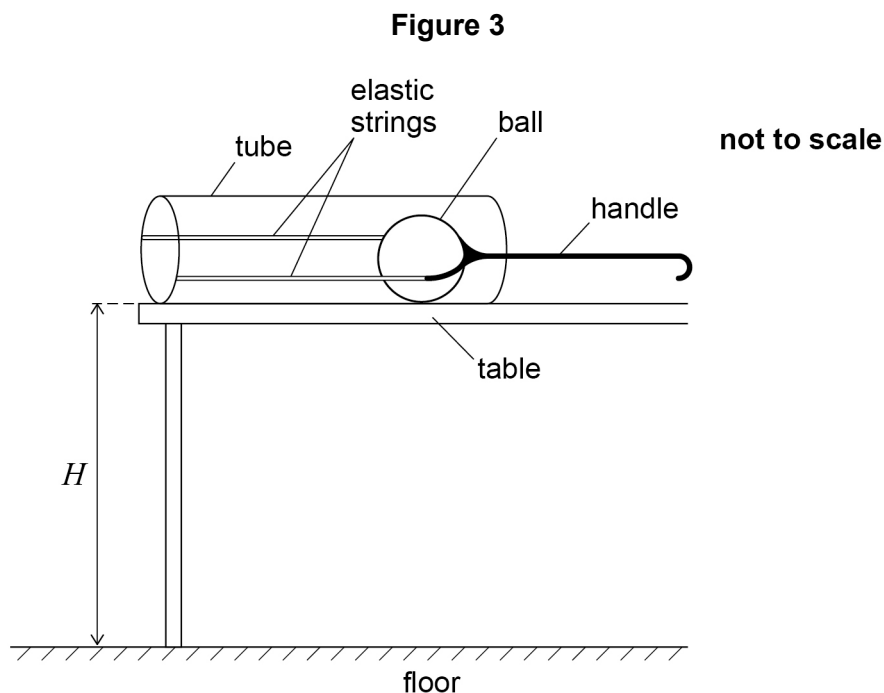
Young modulus = _____ Pa

4



Don't write

Figure 3 shows apparatus that is used to investigate projectile motion. The apparatus is placed on a table at height H above the horizontal floor.



A ball of mass m and two elastic strings are inside an open tube.
A handle is pulled back to stretch the elastic strings. When the handle is released, the ball is launched horizontally from the tube.
The ball leaves the tube with kinetic energy E .

The ball travels a horizontal distance D between leaving the tube and hitting the floor.

Show that

$$D = 2\sqrt{\frac{HE}{mg}}$$

[3 marks]



0 7 . 2

State **one** assumption you made in your answer to Question 07.1.

[1 mark]

Question 7 continues on the next page



07.3

When the handle is pulled back, the extension in each string is ΔL .
When the handle is released, 50% of the stored elastic energy is transferred into kinetic energy of the ball.

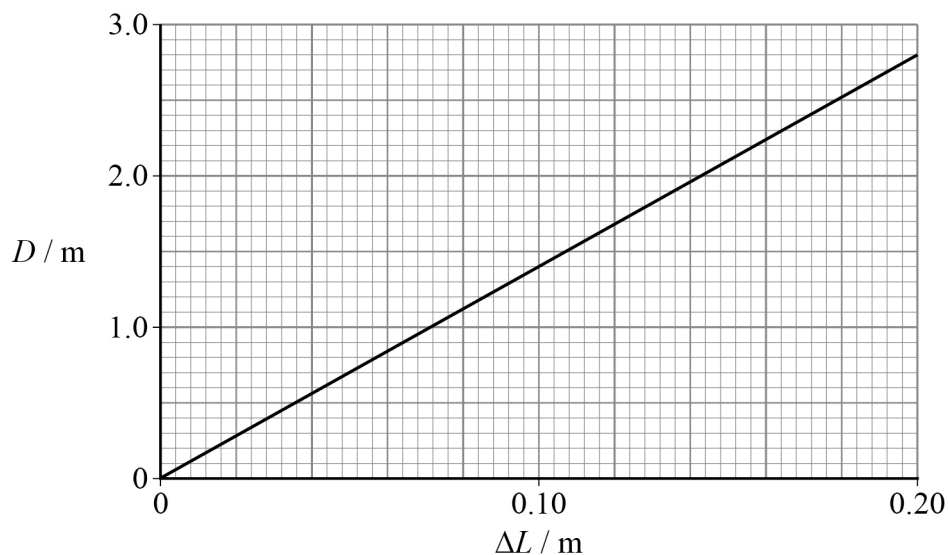
The relationship between D and ΔL is

$$D = \Delta L \sqrt{\frac{2Hk}{mg}}$$

where k is the spring constant of each string.

Figure 4 shows the variation of D with ΔL .

Figure 4



Determine k .

$$m = 30 \text{ g}$$

$$H = 0.80 \text{ m}$$

[3 marks]

$$k = \underline{\hspace{10em}} \text{ N m}^{-1}$$

7

Turn over for the next question

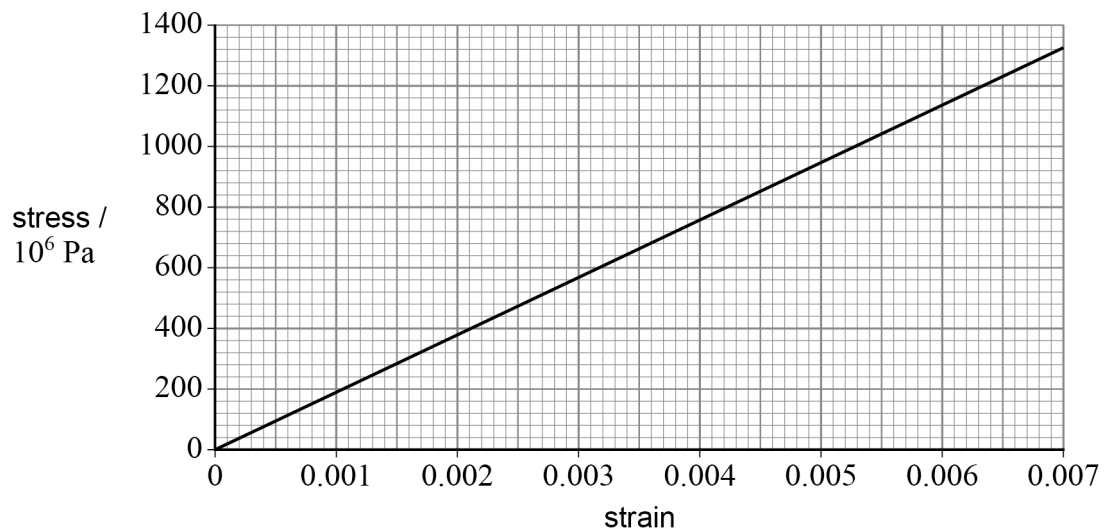
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Figure 11 shows the variation of stress with strain for a type of steel.

Figure 11



09.1 Show that the Young modulus of the steel is approximately 200 GPa.

[1 mark]

Question 9 continues on the next page

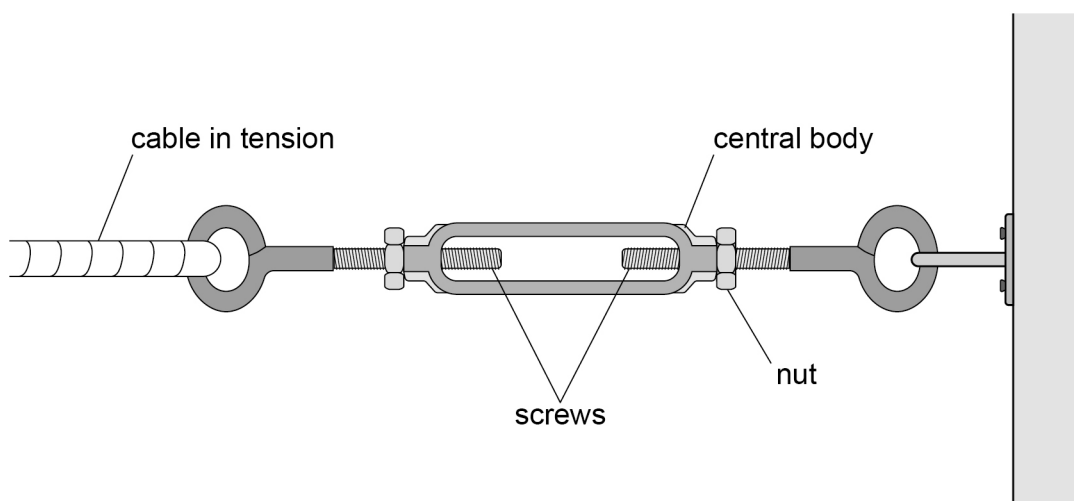
Turn over ►



A length of steel cable is attached to a wall using a device called a turnbuckle. The other end of the cable is also fixed.

Figure 12 shows the turnbuckle.

Figure 12



The turnbuckle is used to adjust the tension in the cable. As the central body is rotated, the separation of the screws changes. This change depends on the direction of rotation and the amount of rotation of the central body.

0 9 . 2

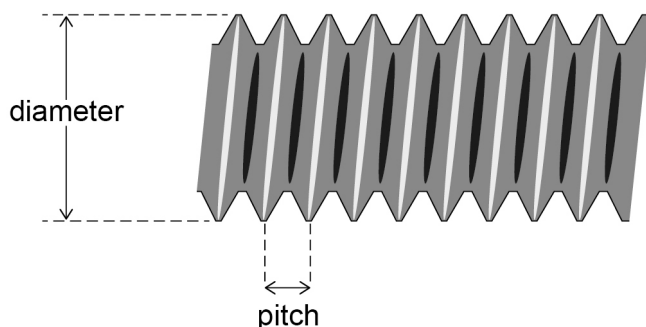
State and explain what must be done to the turnbuckle to increase the tension in the cable.

[1 mark]



Figure 13 shows a section of one screw from the turnbuckle.
With one complete rotation of the central body, both screws move through a distance of one pitch relative to the central body.

Figure 13



Both screws have the same diameter and the same pitch.

The diameter of each screw is 14 mm.

0 9 . 3 For one complete rotation of the central body, the extension of the cable is 1.3 mm.

Determine the ratio $\frac{\text{diameter}}{\text{pitch}}$.

[1 mark]

$$\frac{\text{diameter}}{\text{pitch}} = \underline{\hspace{2cm}}$$

The cable is initially under tension with a stress of 210 MPa. The unstretched length of the cable is 1.8 m.

0 9 . 4 Determine, using **Figure 11**, the initial strain in the cable.

[1 mark]

$$\text{strain} = \underline{\hspace{2cm}}$$

Question 9 continues on the next page

Turn over ►



The central body is now turned through eight complete rotations.

0 9 . 5 Calculate the new strain in the cable.

[2 marks]

strain = _____

0 9 . 6 The cross-sectional area of the cable is $3.7 \times 10^{-5} \text{ m}^2$.

Calculate the new tension in the cable.

[3 marks]

tension = _____ N



Turn over for the next question

*Do not write
outside the
box*

**DO NOT WRITE ON THIS PAGE
ANSWER IN THE SPACES PROVIDED**

Turn over ►



Don't write

A student investigates the friction on a block that slides across a horizontal surface.

The student places the block onto the surface as shown in **Figure 14**.

The student compresses the spring by moving the block a distance x to the left, as shown in **Figure 15**.

When the student releases the block, it accelerates across the surface. Friction then causes the block to decelerate and come to rest as shown in **Figure 16**.

Figure 14

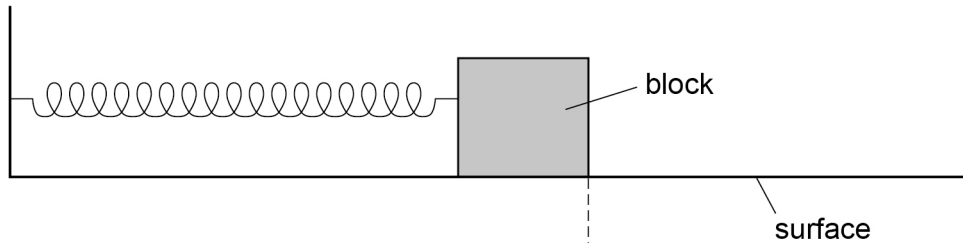


Figure 15

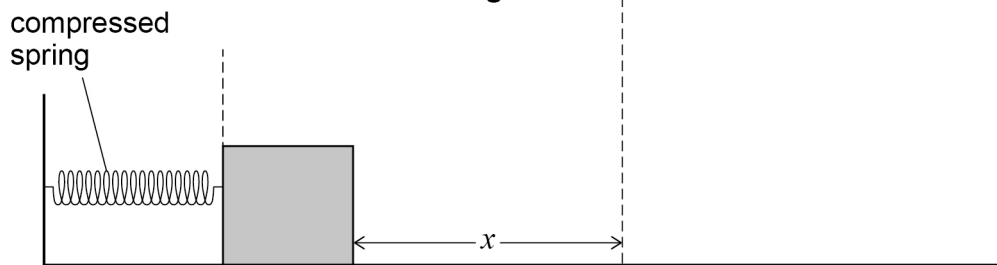
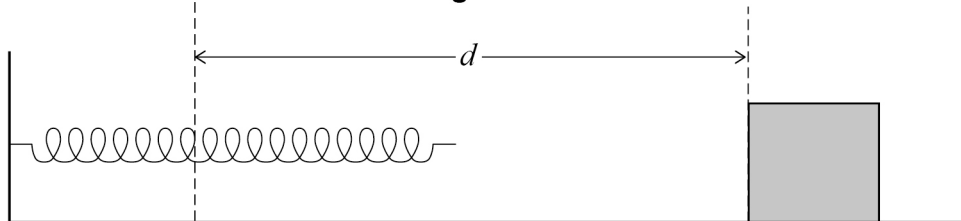


Figure 16



The student makes repeated measurements of distance d .

Table 1 shows the recorded values of d for one particular value of x .

Table 1

d / mm	79	82	81	79	80
-----------------	----	----	----	----	----

1 0 . 1

Calculate the percentage uncertainty in the mean value of d .

[2 marks]

percentage uncertainty = _____

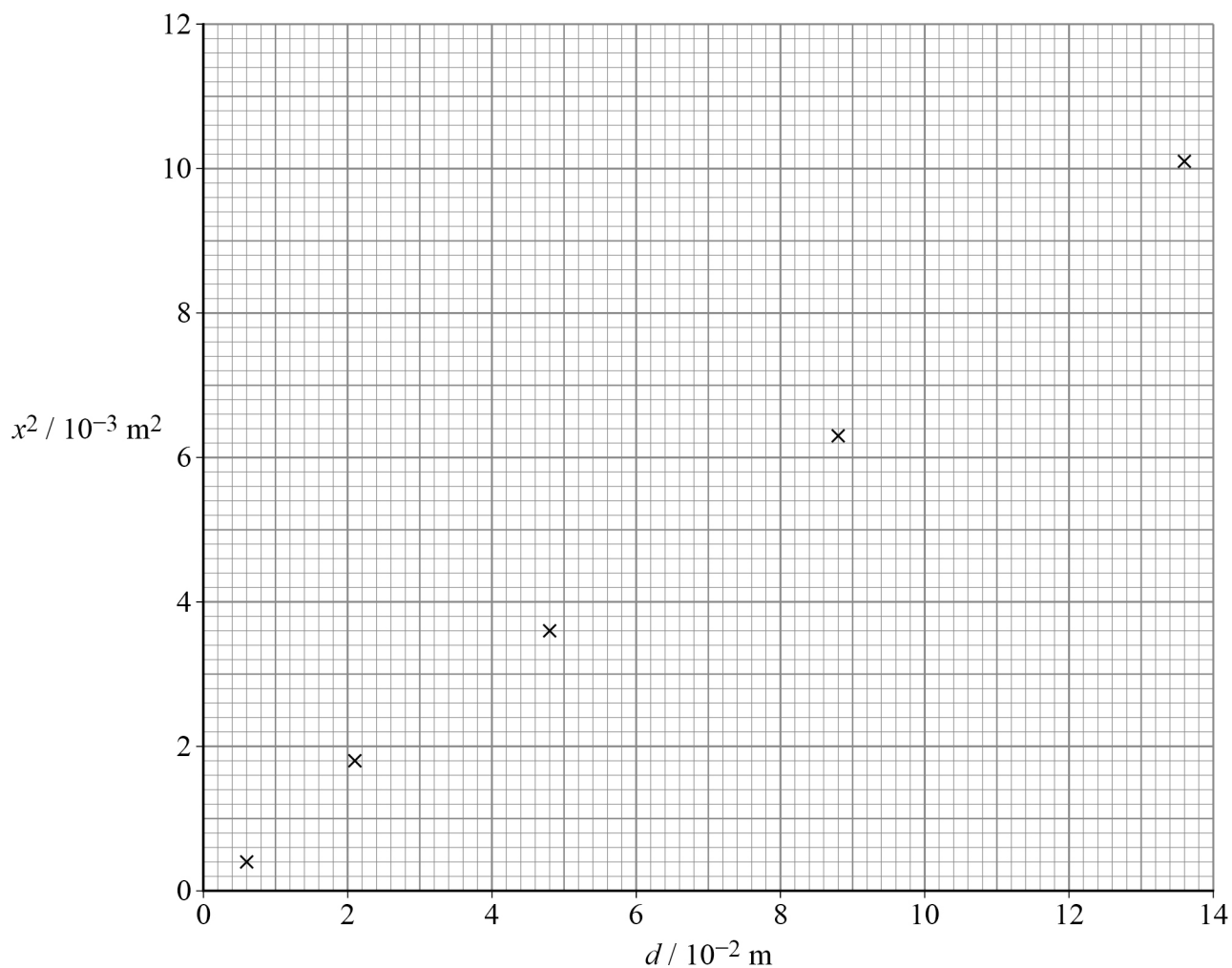
Question 10 continues on the next page



The student determines mean values of d for a range of values of x .

Figure 17 shows the student's plot of x^2 against d .

Figure 17



1 0 . 2 Draw a best-fit line on **Figure 17**.

[1 mark]



The frictional force F between the surface and the block is related to the weight W of the block by the equation

$$F = \mu W$$

where μ is a constant.

The elastic potential energy E in the compressed spring is given by

$$E = \frac{1}{2} kx^2$$

where k is the spring constant of the spring.

1	0
---	---

 .

3

 Show that $x^2 = \frac{2\mu mgd}{k}$

where m is the mass of the block.

[1 mark]

1	0
---	---

 .

4

 Determine μ using **Figure 17**.

$$k = 60 \text{ N m}^{-1}$$

$$m = 0.75 \text{ kg}$$

[3 marks]

$$\mu = \underline{\hspace{2cm}}$$

END OF SECTION B



Section C

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For each question select the best response.

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CORRECT METHOD



WRONG METHODS



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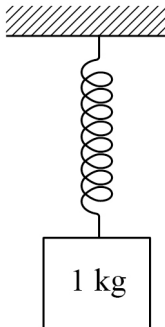
You may do your working in the blank space around each question but this will not be marked.
Do **not** use additional pages for this working.

Arrangements of springs are used to suspend different masses.
All the springs are identical.

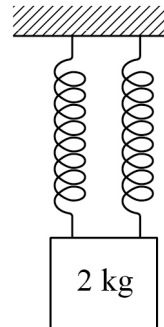
Which arrangement of springs has the greatest extension?

[1 mark]

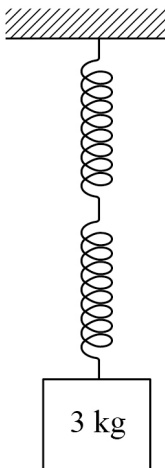
A



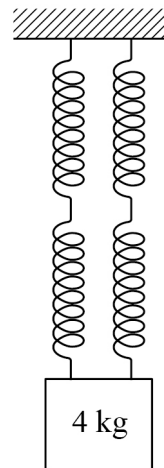
B



C



D



A

☐

B

☐

C

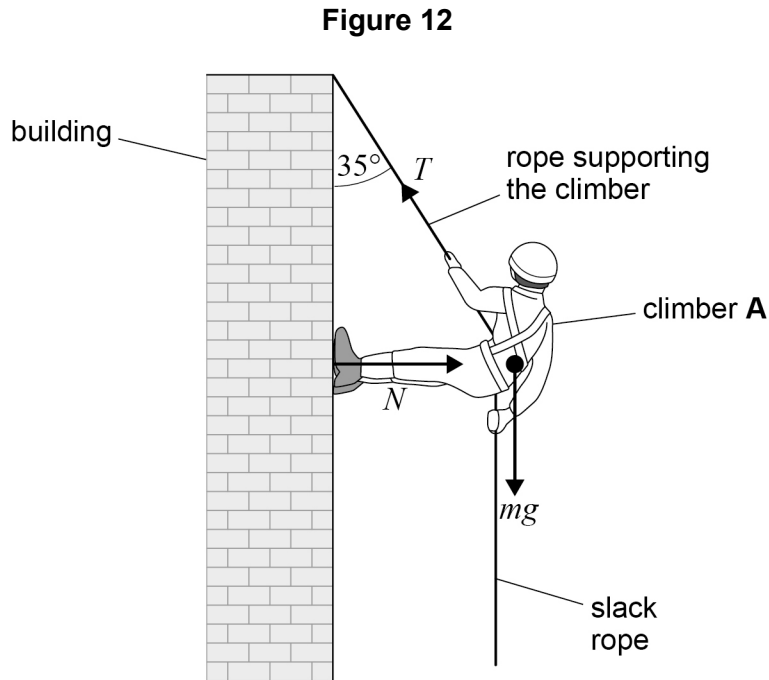
☐

D

☐

Don't write

Figure 12 shows a stationary climber **A** of weight 750 N on the vertical side of a building. **A** is supported by a rope and his legs are perpendicular to the building.



The section of the rope supporting **A** is in tension T and makes an angle of 35° to the vertical.

A normal contact force N acts on **A**'s feet where they touch the building.
The weight of the rope is negligible.

The section of the rope supporting **A** has an unstretched length of 1.80 m.
The extension of the rope is 40 mm.

The Young modulus of the rope material is 260 MPa.

Calculate the diameter of the rope.

[4 marks]

diameter = _____ mm



1 1 . 2

A moves down the rope and then stops lower down the building.
His legs are again perpendicular to the building.

The magnitudes of T and N are different from their magnitudes when **A** is in the position shown in **Figure 12**.

State and explain the differences.

[3 marks]

T _____

N _____

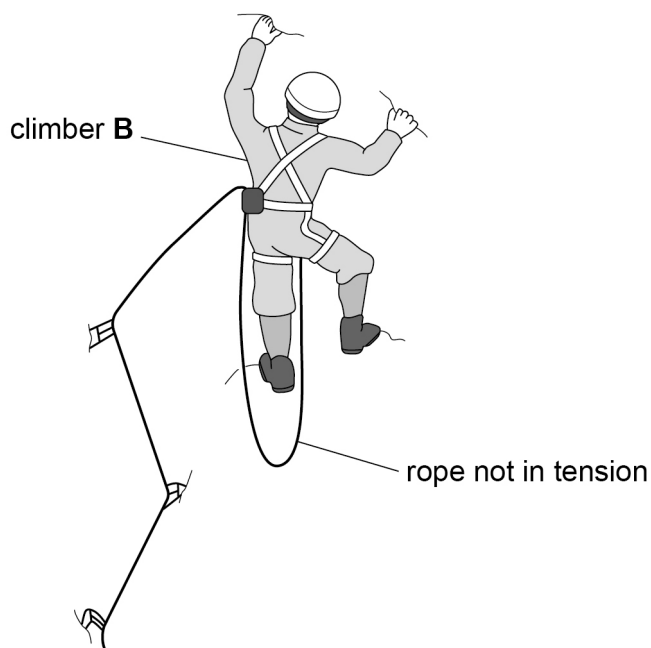
Question 11 continues on the next page



1 1 . 3

Figure 13 shows climber **B** who is climbing up a cliff. **B** is attached to a safety rope that is not in tension. The bottom end of the safety rope is fixed in place.

Figure 13



If **B** falls, the rope stops her after a short distance.

The ropes used in **Figure 12** and **Figure 13** have the same diameter but are made from different materials.

For safety reasons, the rope in **Figure 13** has a much lower stiffness than the rope in **Figure 12**.

Explain why.

[2 marks]

END OF SECTION B



Section C

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CORRECT METHOD



WRONG METHODS



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You may do your working in the blank space around each question but this will not be marked.
Do **not** use additional pages for this working.

A tensile force acts on a metal wire of initial length L and cross-sectional area A . This force causes an extension ΔL . The Young modulus of the metal is E .

What is the stiffness k of the wire?

[1 mark]

A $\frac{EA}{L}$ ☐

B $\frac{EL}{A}$ ☐

C $\frac{EL}{A(\Delta L)^2}$ ☐

D $\frac{E(\Delta L)^2}{AL}$ ☐



Don't write

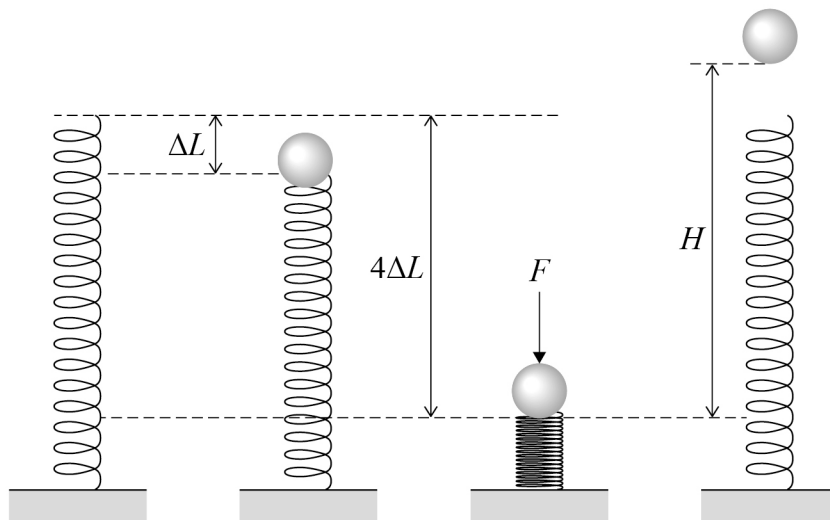
A light spring is placed on the floor and is in equilibrium.

A ball of mass 130 g is placed on the top of the spring, causing the spring to compress by a distance ΔL .

A force F is then applied to the top of the ball so that the spring compression increases to $4\Delta L$.

F is removed, causing the ball to move vertically through a distance H .

The spring has a spring constant of 58 N m^{-1} and obeys Hooke's law.



What is the maximum value of H ?

[1 mark]

A 0.044 m

☐

B 0.088 m

☐

C 0.18 m

☐

D 0.35 m

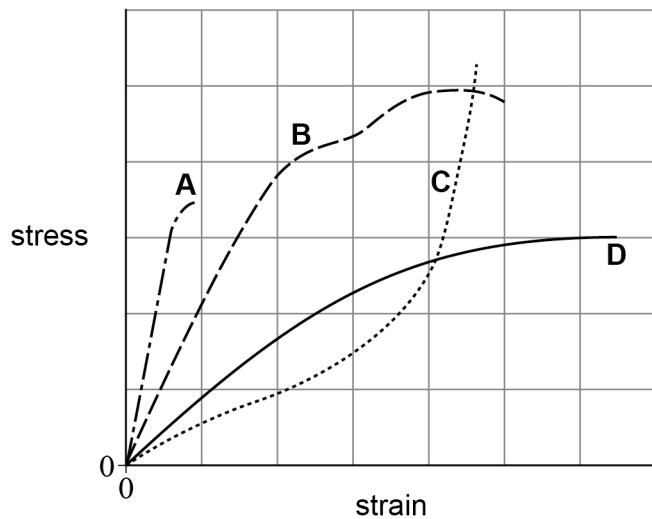
☐


Don't write

2	0
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Which stress–strain curve shows the greatest work done per unit volume?

[1 mark]



A



B



C



D



A student measures the diameter and the mass of a solid metal ball. She uses her values to calculate the density of the metal.

The student quotes the density as:

$$\rho = 8000 \pm 120 \text{ kg m}^{-3}$$

The uncertainty in the measurement of the diameter of the ball is 0.4%.

What is the uncertainty in the measurement of the mass?

[1 mark]

A 0.3%

☐

B 0.7%

☐

C 1.1%

☐

D 1.5%

☐

Don't write

2**2***Do not write
outside the
box*

The radius R_a of a gold atom is approximately 140 pm.

The radius R_n of the nucleus of the gold atom is approximately 7.0 fm.

What is $\frac{R_a}{R_n}$?

[1 mark]

A 5.0×10^{-8}

☐

B 5.0×10^{-5}

☐

C 2.0×10^4

☐

D 2.0×10^7

☐

0 1 . 1

Describe plastic behaviour in a material.

[1 mark]

0 1 . 2

Describe the behaviour of a brittle material.

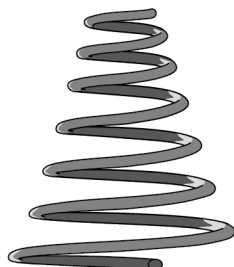
[1 mark]

fundamental (base) units of stiffness = _____

Figure 1 shows a spring **S** at its natural length. **S** does not obey Hooke's law.

Figure 2 shows **S** when it is compressed.

Figure 1



natural length

Figure 2



compressed

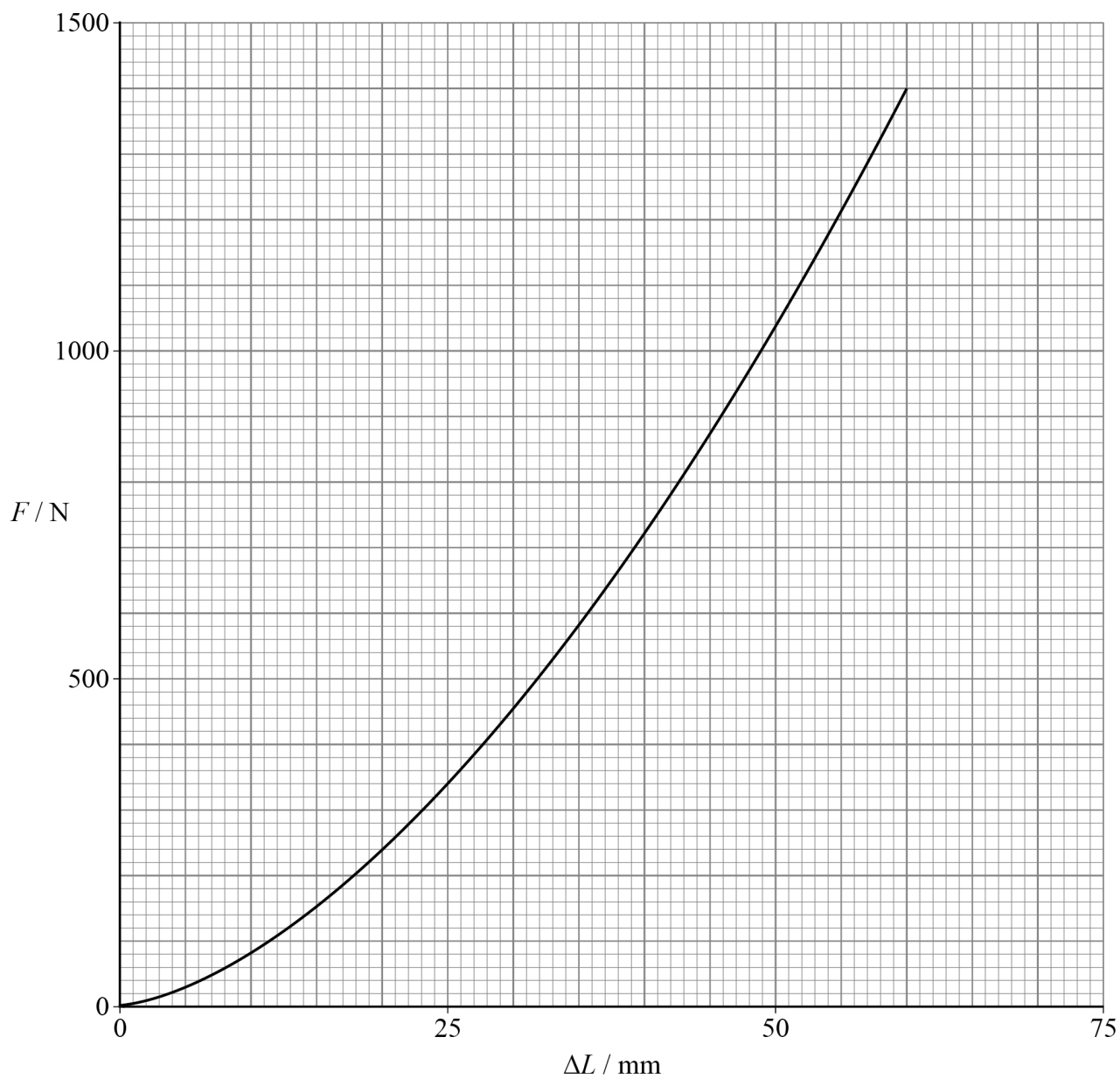
Question 3 continues on the next page



A force F is applied to compress **S**.
 F and the change in length ΔL of **S** are measured.

Figure 3 shows the variation of ΔL with F .

Figure 3



0 3 . 2 **S** is compressed until $F = 1400 \text{ N}$.

Determine the work done.

[3 marks]

work done = _____ J

0 3 . 3 Explain how **Figure 3** shows how the stiffness of **S** changes as **S** is compressed.

[2 marks]

Turn over for the next question

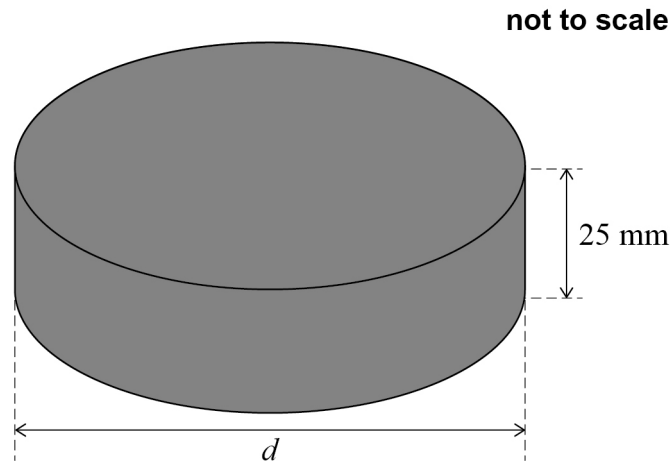


Don't write

This question is about a procedure to test the friction of an ice surface where a game will be played.

Figure 9 shows a rubber disc used in the procedure.

Figure 9



The mass of the disc is 0.17 kg.

The density of the rubber is 1500 kg m^{-3} .

The height of the disc is 25 mm and the diameter is d .

Calculate d in mm.

[3 marks]

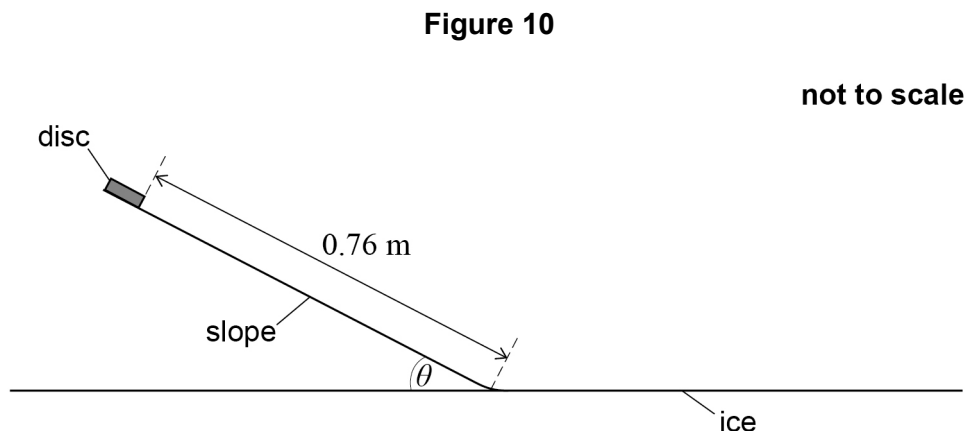
$d =$ _____ mm



The procedure to test the friction of an ice surface measures the distance that the disc will travel across a horizontal ice surface for a given initial speed.

The disc is placed at the top of a slope. The angle between the ice and the slope is increased. The disc starts to slide down the slope when the angle is θ .

Figure 10 shows the disc at the instant that it starts to slide.



The distance that the disc travels down the slope is 0.76 m.

A constant frictional force of 0.41 N acts on the disc as it moves down the slope.

The slope is designed so that the disc always reaches the end of the slope at a speed of 2.1 m s^{-1} .

0 8 . 2 Calculate θ .

[4 marks]

$\theta =$ _____ °

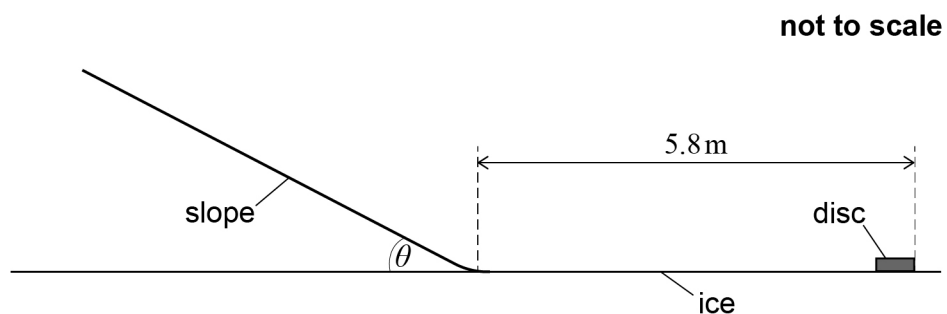
Question 8 continues on the next page

Turn over ►



Figure 11 shows the disc when it has travelled across the ice and come to rest. The horizontal distance travelled by the disc along the ice is measured as 5.8 m.

Figure 11



0 8 . 3

The deceleration of the disc is uniform.

The initial speed of the disc on the ice is 2.1 m s^{-1} .

Calculate the time taken for the disc to travel along the ice before coming to rest.

[2 marks]

time = _____ s



0 8 . 4

The procedure tests the friction between the ice and the disc for a game played on ice.

The procedure is repeated three times in the same location and then done in different locations across the area where the game is to be played.

Explain why the measurement is:

- repeated three times in the same location
- done in different locations.

[2 marks]

11

END OF SECTION A



Section B

Answer **all** questions in this section.

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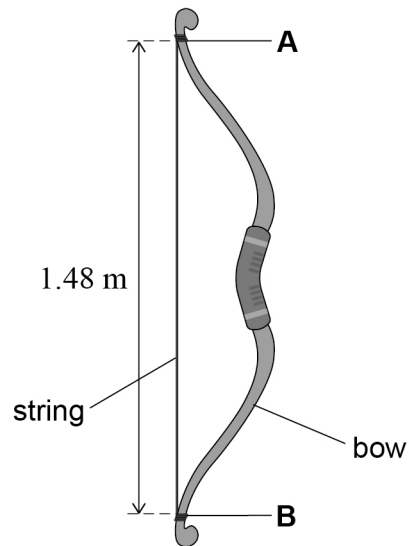
Figure 12 shows a bow that is used to shoot an arrow at a target.

A string is attached to the bow at points **A** and **B**.

The string is in tension.

The string obeys Hooke's law.

Figure 12



The distance between **A** and **B** is 1.48 m.

The tension in the string is 450 N and the tensile strain is 0.072

The diameter of the string is 1.64 mm.

10.1

Show that the Young modulus of the material of the string is approximately 3.0 GPa.

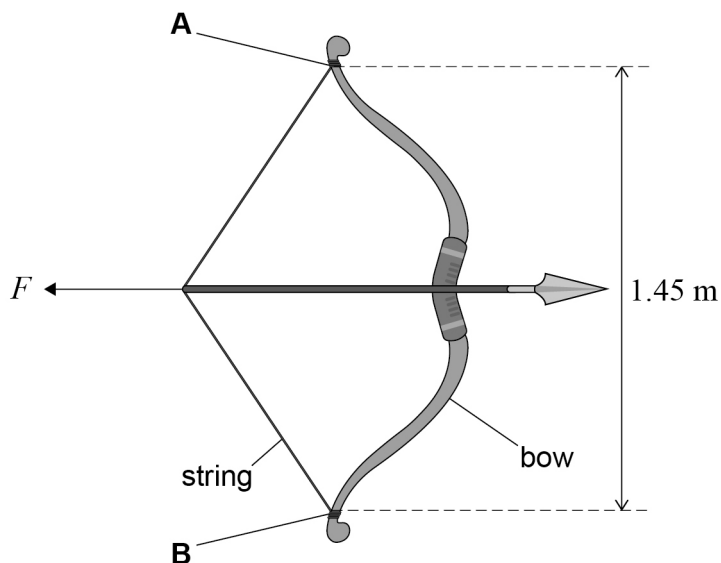
[3 marks]



An arrow is placed in the bow and a force F is applied to pull the string.

Figure 13 shows the string and arrow in a new equilibrium position.

Figure 13



1 0 . 2

The distance between **A** and **B** is now 1.45 m.

The tension in the string is now 930 N and the new length of the string is 1.60 m.

Calculate F .

[2 marks]

$F =$ _____ N

Question 10 continues on the next page

1 0 . 3

F is removed so that the bow accelerates the arrow.

Explain how the acceleration of the arrow varies as the length of the string decreases from 1.60 m to 1.48 m.

[3 marks]

8

END OF SECTION B



Section C

Each of the questions in this section is followed by four responses, **A**, **B**, **C** and **D**.

For each question select the best response.

Only **one** answer per question is allowed.

For each question, completely fill in the circle alongside the appropriate answer.

CORRECT METHOD



WRONG METHODS



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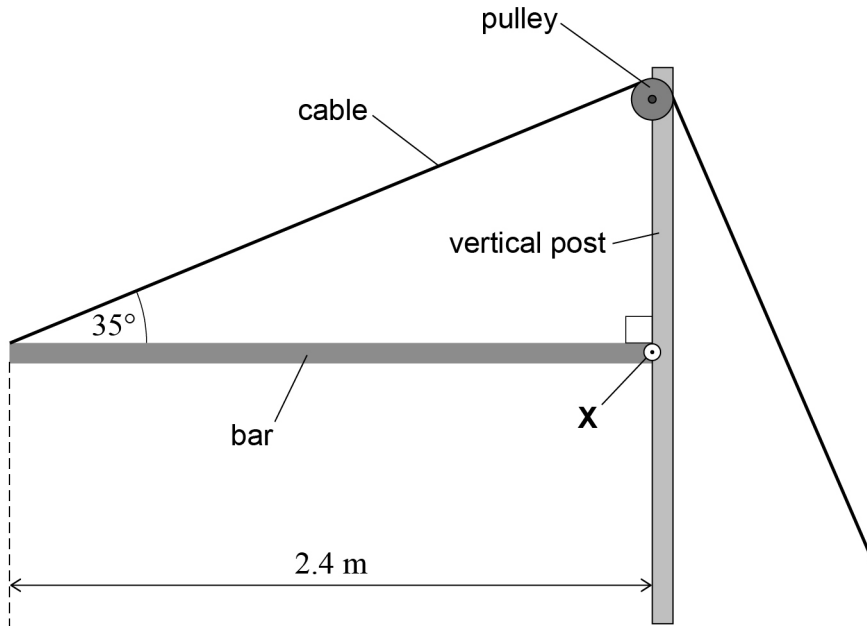
You may do your working in the blank space around each question but this will not be marked.
Do **not** use additional pages for this working.

Figure 7 shows a system used to stop cars from entering a car park.

A uniform horizontal bar of length 2.4 m is attached by a pivot to a vertical post at point **X**.

A cable is attached to the left-hand end of the bar and passes over a pulley. The cable makes an angle of 35° with the bar. The tension in the cable is 230 N.

Figure 7



Calculate the mass of the bar.

[3 marks]

mass = _____ kg



0 9 . 2

The cable has a diameter of 11 mm and an unstretched length of 5.2 m.
The Young modulus of the material used to make the cable is 205 GPa.

Determine the extension of the cable in **Figure 7**.

[2 marks]

extension = _____ m

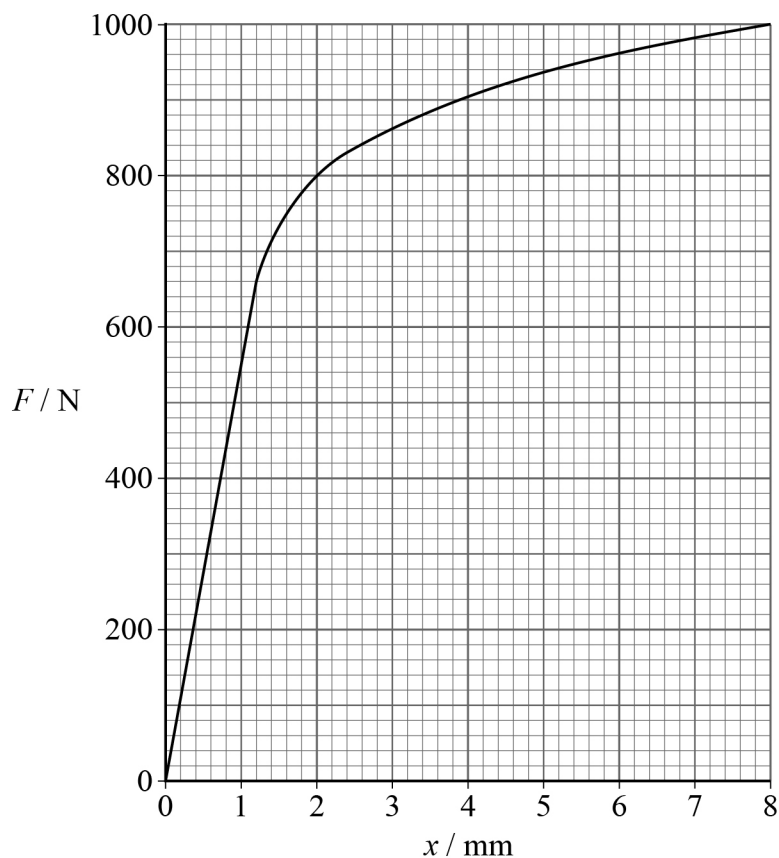
Question 9 continues on the next page



A new cable is tested as a possible replacement for the cable in **Figure 7**.

Figure 8 shows how the extension x varies with the force F applied to the new cable as it is loaded to a maximum of 1000 N.

Figure 8



0 9 . 3 Determine the force applied to the cable at the limit of proportionality.

[1 mark]

force = _____ N



0 9 . 4

The cable is loaded to 1000 N and is then unloaded. The cable is found to have undergone plastic deformation. When the cable is unloaded, some of the work done in stretching the cable cannot be recovered. The amount of work that cannot be recovered is W .

Deduce W using **Figure 8**.

[3 marks]

$W =$ _____ J

9

END OF SECTION A



Section B

Answer **all** questions in this section.

*Do not write
outside the
box*

Figure 9 shows a system consisting of a pair of identical springs and a rod of negligible mass.

The upper end of each spring is fixed. The rod is attached to the bottom of the springs so that the springs are parallel to each another.

The length L_0 of each unstretched spring is measured using a ruler with millimetre divisions.

Figure 9

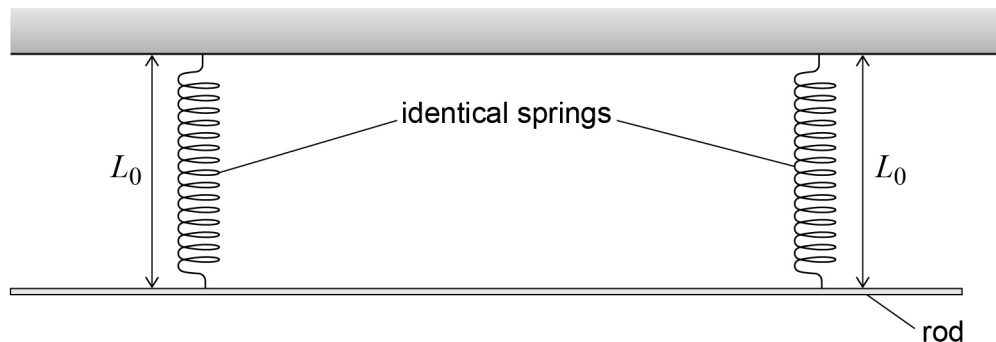
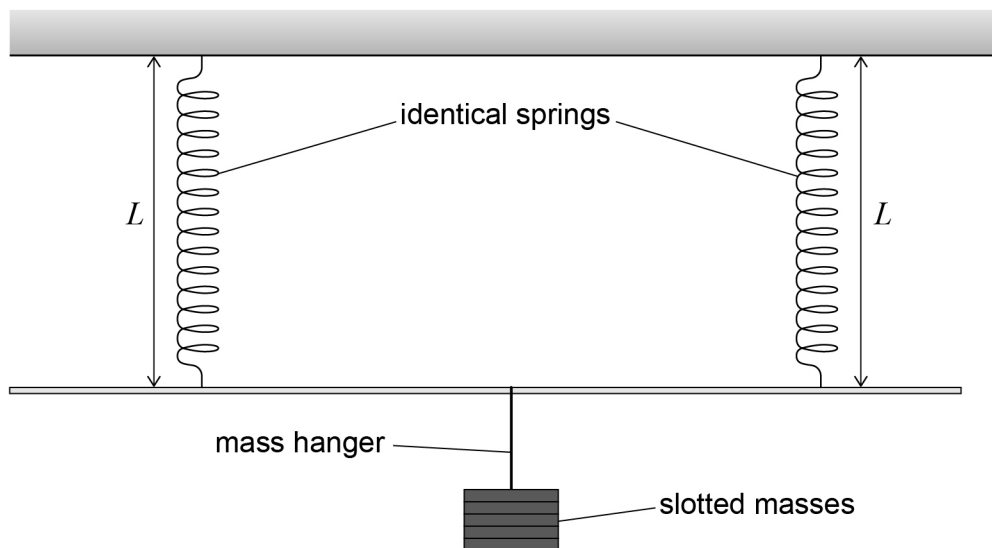


Figure 10 shows a mass hanger and slotted masses suspended from the rod at the mid-point between the springs.

Figure 10



Known masses are added to gradually increase the force F applied to the springs.
For each value of F , the length L is measured and the extension ΔL is calculated.

1 0 . 1

Explain why the minimum uncertainty in ΔL is 2 mm.

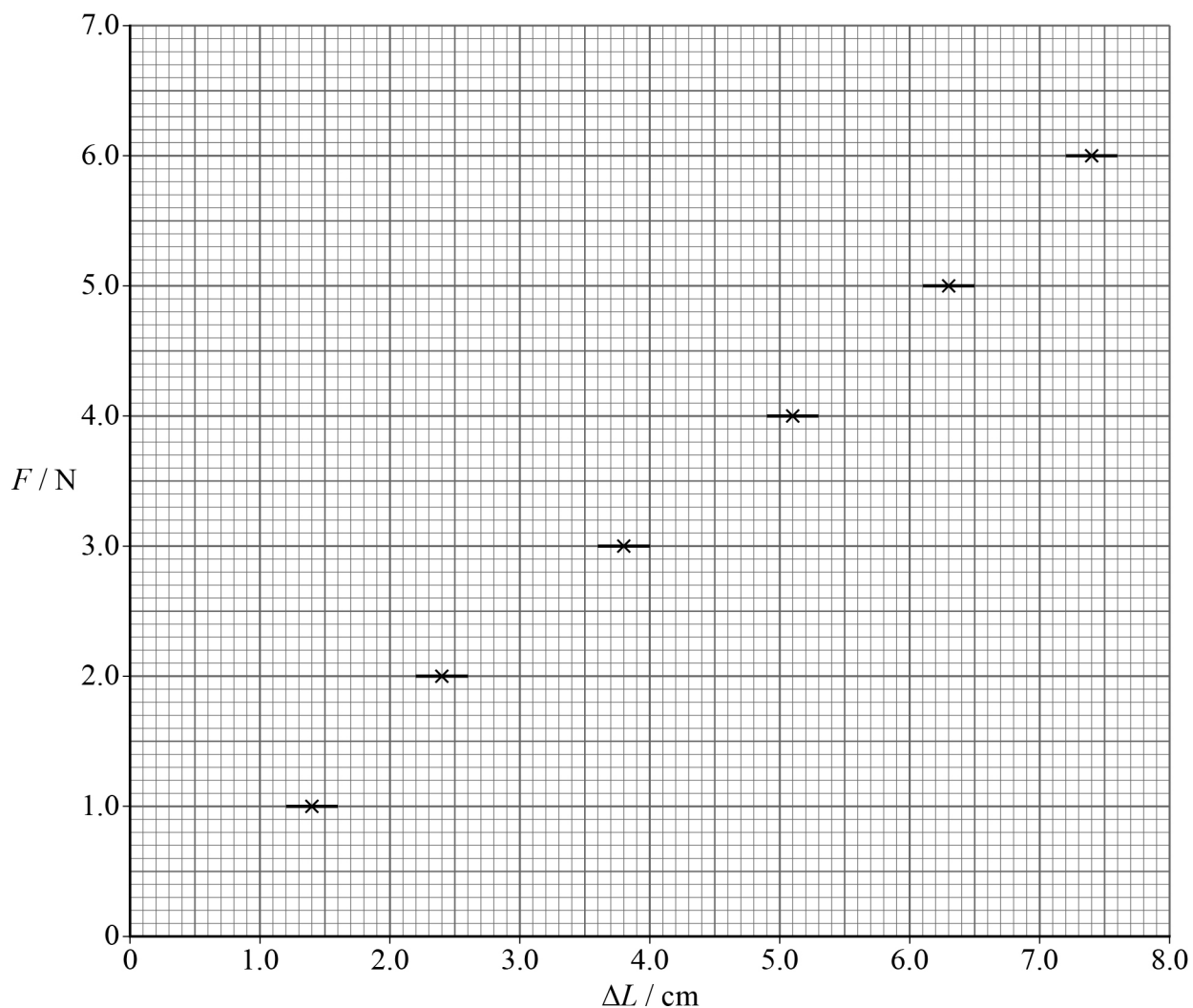
[2 marks]

Question 10 continues on the next page



Figure 11 shows data points with error bars plotted on a graph of F against ΔL .

Figure 11



1 0 . 2 Draw, on **Figure 11**, the two best-fit lines that show the maximum and minimum gradients for the graph.

[1 mark]

k_t is the total spring constant of the system.

1 0 . 3

Use **Figure 11** to determine a value, in N cm^{-1} , for k_t .

[3 marks]

$k_t =$ _____ N cm^{-1}

1 0 . 4

Determine, in N cm^{-1} , the uncertainty in k_t .

[1 mark]

uncertainty = _____ N cm^{-1}

1 0 . 5

The experiment is repeated using a different pair of identical springs. Each new spring has a larger spring constant than each original spring.

State and explain the effect of this change on the value for the uncertainty in k_t .

[2 marks]



Don't write

A metal wire has initial length L and diameter d . An extension e is produced in the wire when a force is applied.

A second wire of the same material has initial length $\frac{L}{2}$ and diameter $\frac{d}{2}$.

What is the extension of the second wire when the same force is applied?

[1 mark]

A $\frac{e}{4}$



B $\frac{e}{2}$



C e



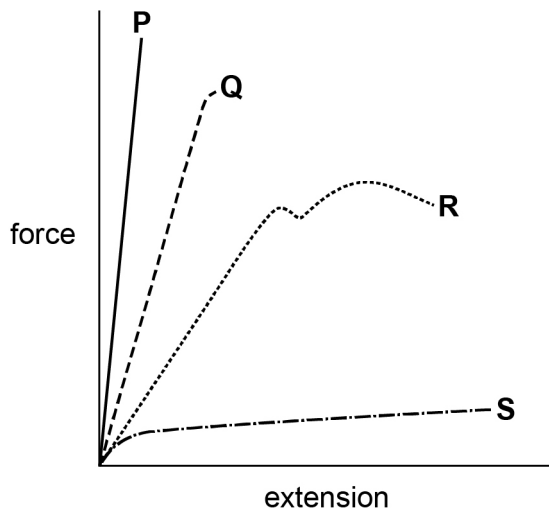
D $2e$



Don't write

2	5
---	---

The graph shows the variation of extension with force for four wires **P**, **Q**, **R** and **S**, up to the points at which they break.



Which is **not** correct?

[1 mark]

A **S** is less brittle than **Q**.

☐

B **P** is brittle.

☐

C **S** has the largest spring constant.

☐

D **R** behaves plastically.

☐





[illegible]

0	4
---	---

A radioactive nucleus decays with the emission of an alpha particle.

Describe the changes that occur in the proton number and the nucleon number of the nucleus.

[2 marks]

Proton number _____

Nucleon number _____

0 5 . 1 Compare the relative penetrating powers of alpha and gamma radiation.

[1 mark]

0 5 . 2 Gamma rays from a point source are travelling towards a detector.
The distance from the source to the detector is changed from 1.0 m to 3.0 m.

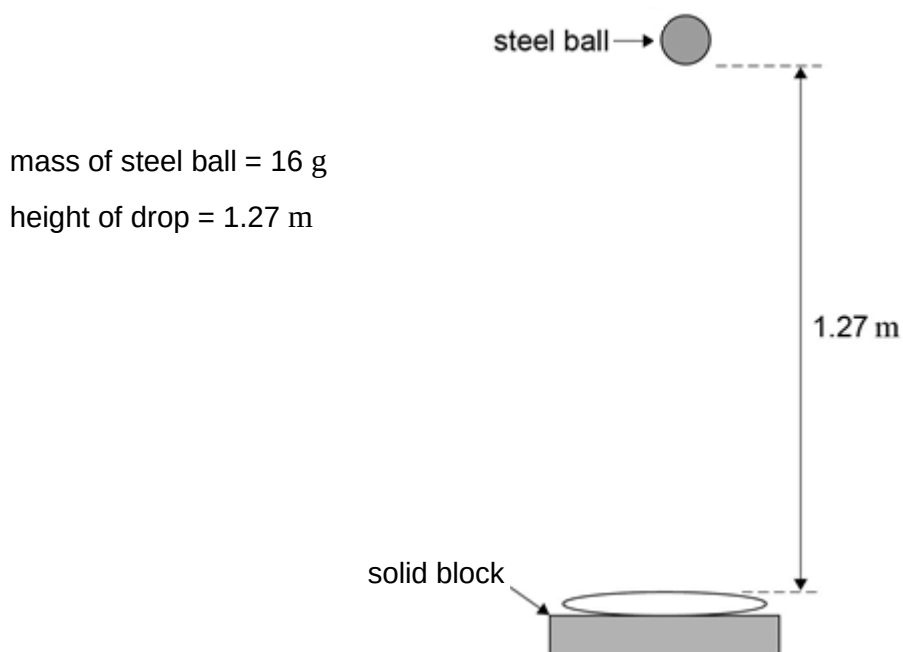
[2 marks]

Calculate the ratio $\frac{\text{intensity of radiation at 3.0 m}}{\text{intensity of radiation at 1.0 m}}$

ratio = _____

0 6 Spectacle lenses may be tested for shock resistance by dropping a small steel ball onto the lens, as shown in **Figure 2**, and then checking the lens for damage.

Figure 2



0 6 . 1 Calculate the speed of the ball just before impact.
Ignore the effects of air resistance in this question.

[2 marks]

speed before impact = _____ m s^{-1}

0 6 . 2 In a test the ball bounced back to a height of 0.85 m.

Calculate the speed of the ball just after impact.

[2 marks]

speed after impact = _____ m s^{-1}

0 6 . 3 Calculate the change in momentum of the ball due to the impact.

[2 marks]

momentum = _____ kg m s^{-1}

Question 6 continues on the next page

0 6 . 4 The time of contact was 40 ms.

Calculate the average force exerted by the ball on the lens during the impact.

[1 mark]

average force = _____ N

0 6 . 5 Explain, with reference to momentum, why the test should also specify the material of the solid block the lens sits on.

[2 marks]

0 7

Metal railway tracks expand when the weather becomes warmer. Therefore, when rails are laid in cold weather they are stretched and fixed into place while still stretched to avoid them deforming. This is called pre-straining.

The following data is typical for a length of steel rail:

Young modulus of steel =	$2.0 \times 10^{11} \text{ Pa}$
cross-sectional area of a length of rail =	$7.5 \times 10^{-3} \text{ m}^2$
amount of pre-strain =	2.5×10^{-5} for each kelvin rise in temperature the rail is expected to experience.

A steel rail is laid when the temperature is 8°C and an engineer decides to use a pre-strain of 3.0×10^{-4} .

0 7

1 Calculate the tensile force required to produce the pre-strain in the rail required by the engineer.

[3 marks]

tensile force = _____ N

0 7

2 Calculate the elastic strain energy stored in a rail of unstressed length 45 m when pre-strained as in Question 7.1.

[2 marks]

elastic strain energy = _____ J

Question 7 continues on the next page

0 7 . 3 Calculate the temperature when there is **no** stress in the steel rail.

[2 marks]

temperature = _____ °C

0 7 . 4 Explain why the engineer does **not** use the highest observed temperature at the location of the railway track to determine the amount of pre-strain to use.

[2 marks]

0 8

A climber falls 2.3 m before being stopped by his climbing rope that is secured above him.

The weight of the climber is 840 N.

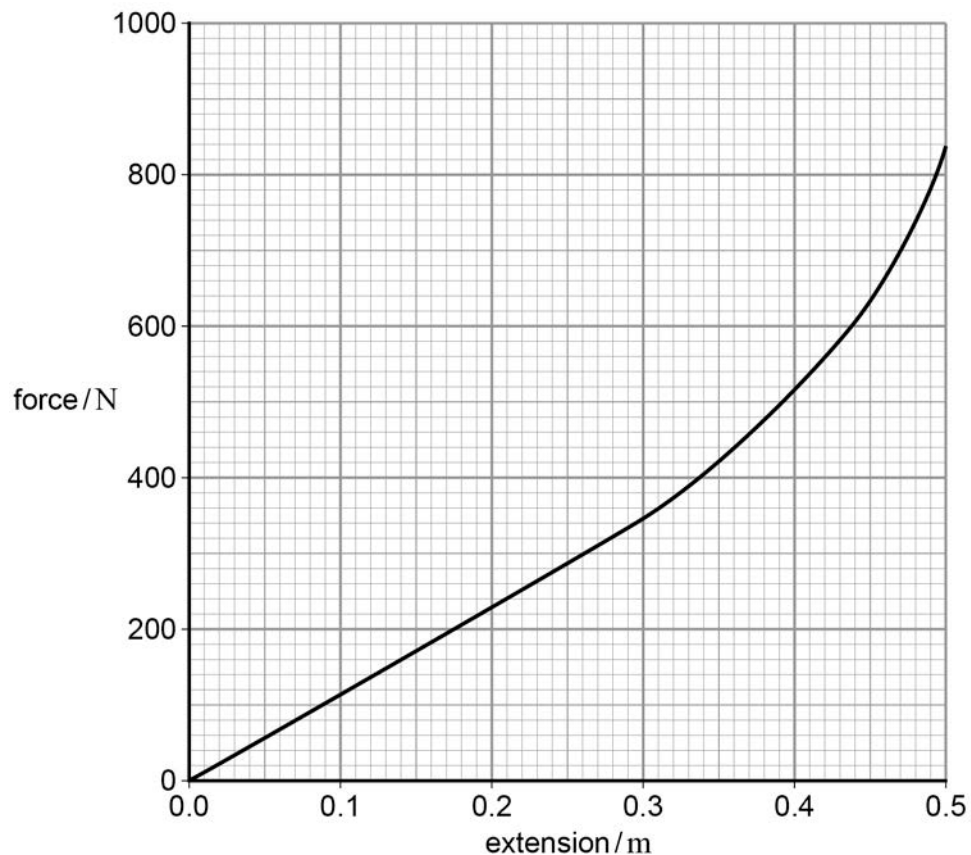
0 8 . 1 Calculate the loss in gravitational potential energy of the climber.

[2 marks]

loss in potential energy = _____ J

Figure 3 shows a force-extension graph for the rope being used.

Figure 3



- 0 8 . 2** Determine the stiffness of the rope when it is being used with forces up to 350 N. Give the appropriate unit.

[2 marks]

stiffness = _____ unit = _____

- 0 8 . 3** Determine the energy stored in the rope when it is stretched by 0.45 m.

[3 marks]

energy = _____ J

09

Table 1 shows information for two isotopes of uranium. The table is incomplete.

Table 1

	number of protons	number of neutrons	specific charge of nucleus /
first isotope	92	143	
second isotope			3.72×10^7

09

. 1

Explain what is meant by isotopes.

[2 marks]

09

. 2

Write the unit for the specific charge in the heading of the last column of **Table 1**.

[1 mark]

09

. 3

Write down the number of protons in the second isotope in **Table 1**.

[1 mark]

09

. 4

Calculate the specific charge of the nucleus of the first isotope.

[3 marks]

specific charge = _____

0	9
---	---

 .

5

 Calculate the number of neutrons in the nucleus of the second isotope.

[3 marks]

number of neutrons = _____

Turn over for the next question

Section B

Answer **all** questions in this section.

1 0

Table 2 gives the values for the activity of a radioactive isotope over a time interval of several seconds.

Table 2

time / s	0	60	120	180	240	300
activity / Bq	480	366	280	214	163	124

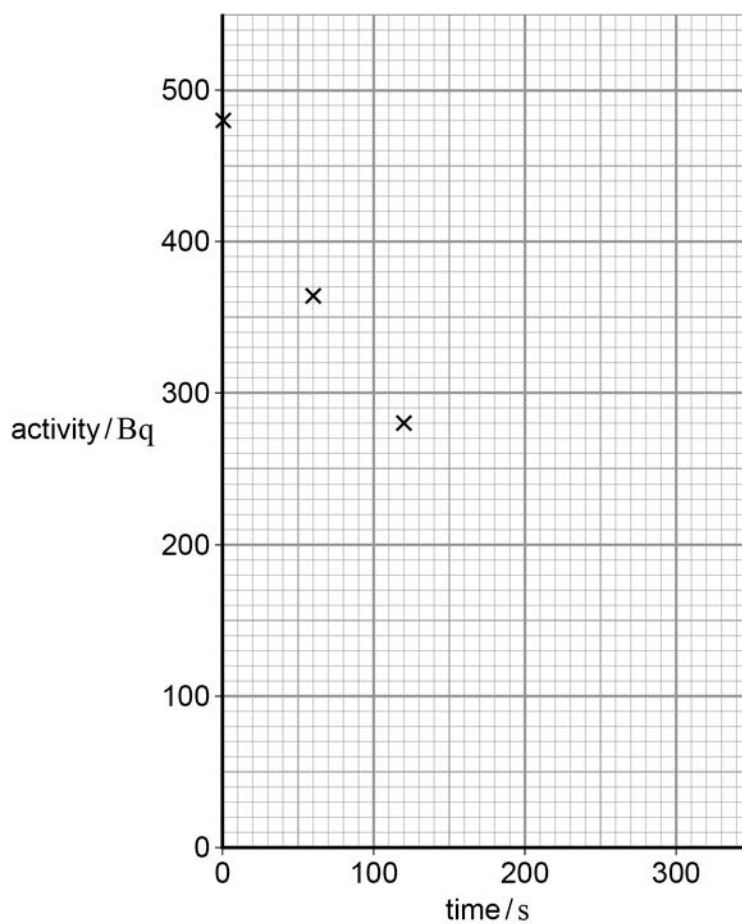
1 0

. 1

Complete the graph in **Figure 4** by plotting the remaining points and drawing an appropriate curve.

[2 marks]

Figure 4



10 . **2** Use the graph to determine the half-life of the isotope.

[2 marks]

half-life = _____ s

Turn over for the next question

Data analysis question

1 1

Capillary action can cause a liquid to rise up a hollow tube. **Figure 5** shows water that has risen to a height h in a narrow glass tube because of capillary action.

Figure 5

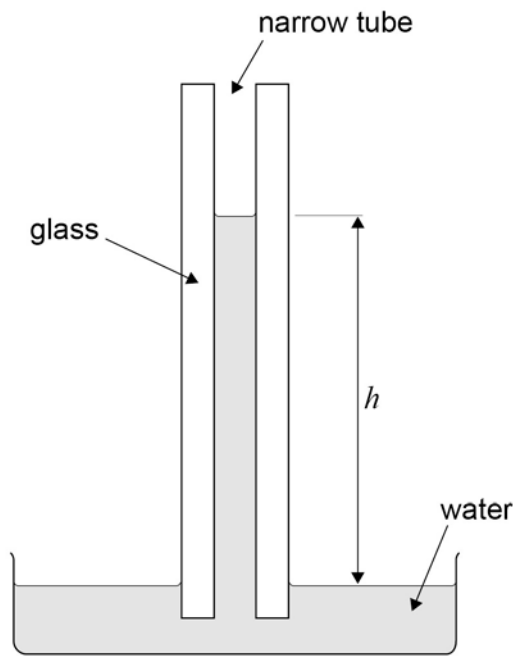
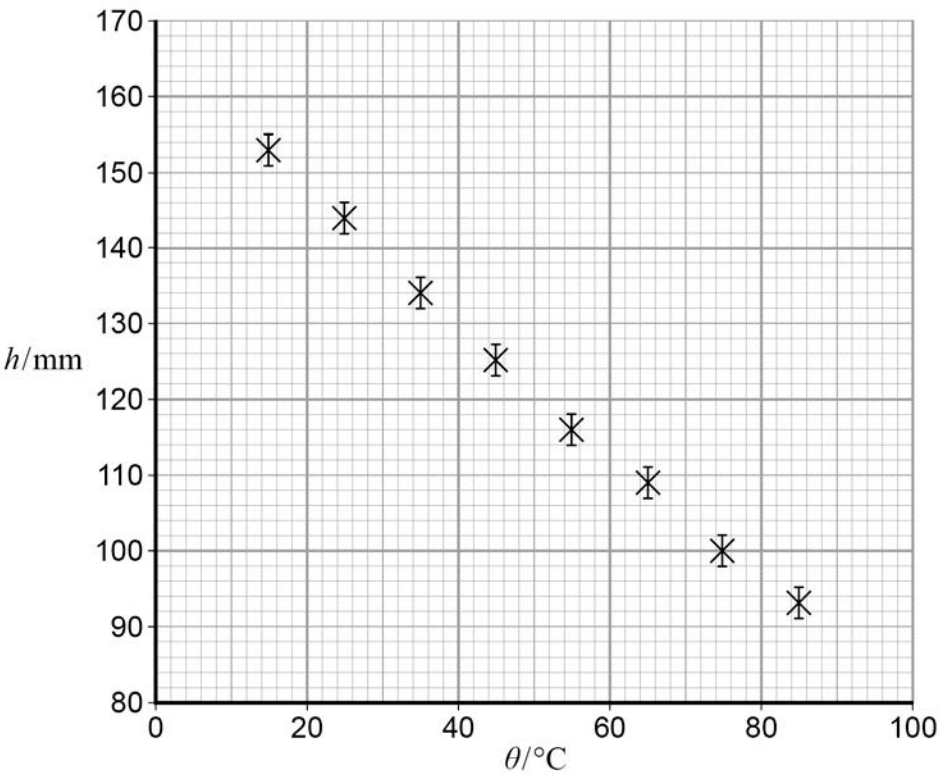


Figure 6 shows the variation of h with temperature θ for this particular tube.

Figure 6



The uncertainty in the measurement of h is shown by the error bars. Uncertainties in the measurements of temperature are negligible.

1 1 . 1 Draw a best-fit line for these data (**Figure 6**).

[2 marks]

1 1 . 2 It is suggested that the relationship between h and θ is

$$h = h_0 - (h_0 k) \theta$$

where h_0 and k are constants.

Determine h_0 .

[2 marks]

$h_0 =$ _____ mm

1 1 . 3 Show that the value of $h_0 k$ is about 0.9 mm K^{-1} .

[3 marks]

1 1 . 4 Determine k . State a unit for your answer.

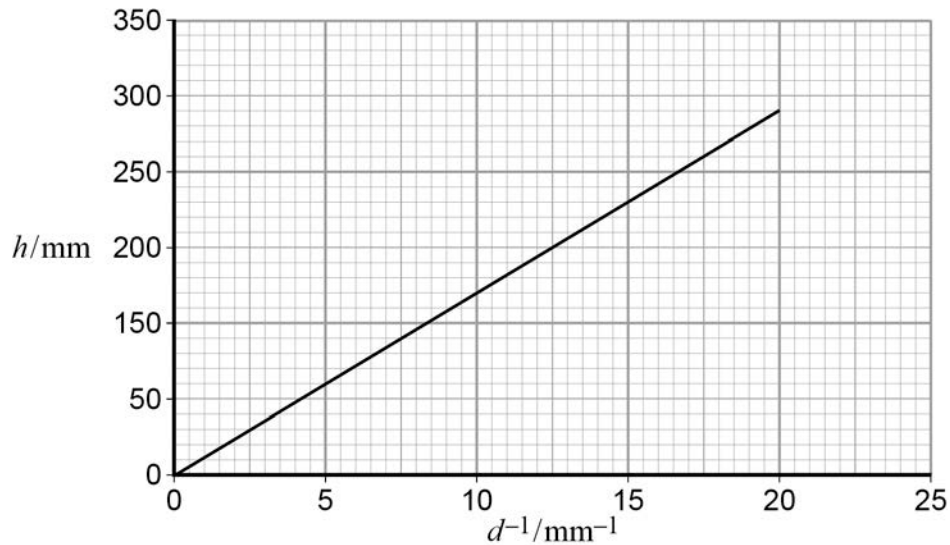
[2 marks]

$k =$ _____ unit = _____

1 1 . 5

A similar experiment is carried out at constant temperature with tubes of varying internal diameter d . **Figure 7** shows the variation of h with $\frac{1}{d}$ at a constant temperature.

Figure 7



It is suggested that capillary action moves water from the roots of a tree to its leaves.

The gradient of **Figure 7** is 14.5 mm^2 .

The distance from the roots to the leaves at the top of the tree is 8.0 m.

Calculate the internal diameter of the tubes required to move water from the roots to the leaves at the top of the tree by capillary action.

[2 marks]

internal diameter = _____ mm

1 1 . 6

Comment on the accuracy of your answer for the internal tube diameter in Question **11.5**.

[1 mark]

Section C

Each of the questions in this section is followed by four responses, **A**, **B**, **C**, and **D**. For each question select the best response.

Only **one** answer per question is allowed.

For each answer completely fill in the circle alongside the appropriate answer.

CORRECT METHOD



WRONG METHODS



If you want to change your answer you must cross out your original answer as shown.



If you wish to return to an answer previously crossed out, ring the answer you now wish to select as shown.



1 2

Which of the following is a scalar quantity?

[1 mark]

A force

☐

B kinetic energy

☐

C momentum

☐

D velocity

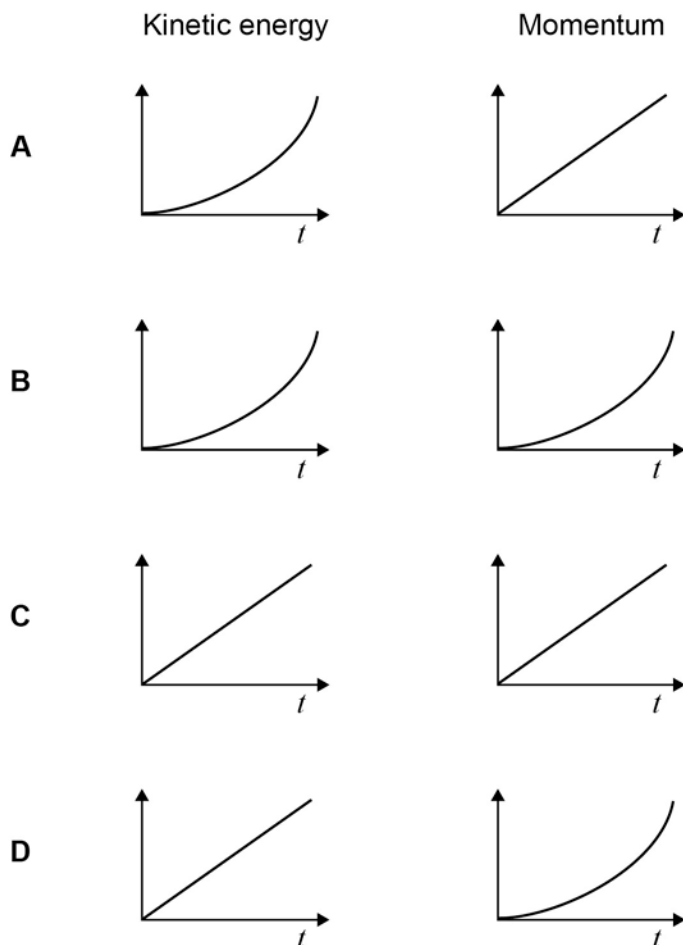
☐

Turn over for the next question

1 3

An object is accelerated from rest by a constant force F for a time t . Which graphs represent the variation of time with the change in the kinetic energy and the change in momentum of the object?

[1 mark]



- A ☐
- B ☐
- C ☐
- D ☐

1 4

An object is dropped from a cliff. How far does the object fall in the third second? Assume that $g = 10 \text{ m s}^{-2}$.

[1 mark]

- A 10 m ☐
- B 20 m ☐
- C 25 m ☐
- D 45 m ☐

1 5

A body falls freely, with negligible air resistance. Which quantity is its rate of change of momentum?

[1 mark]

- A kinetic energy ☐
- B mass ☐
- C power ☐
- D weight ☐

1 6

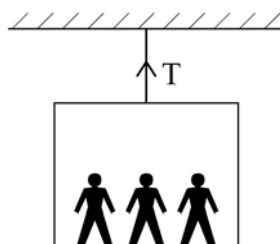
A firework rocket is fired vertically into the air and explodes at its highest point. What are the changes to the total kinetic energy of the rocket and the total momentum of the rocket as a result of the explosion?

[1 mark]

	total kinetic energy of rocket	total momentum of rocket	
A	unchanged	unchanged	☐
B	unchanged	increased	☐
C	increased	unchanged	☐
D	increased	increased	☐

1 7

A lift and its passengers with a total mass of 500 kg accelerates upwards at 2 m s^{-2} as shown. Assume that $g = 10 \text{ m s}^{-2}$.



What is the tension in the cable?

[1 mark]

- A 1000 N ☐
- B 4000 N ☐
- C 5000 N ☐
- D 6000 N ☐

1	8
---	---

Which of the following is **not** a unit of power?

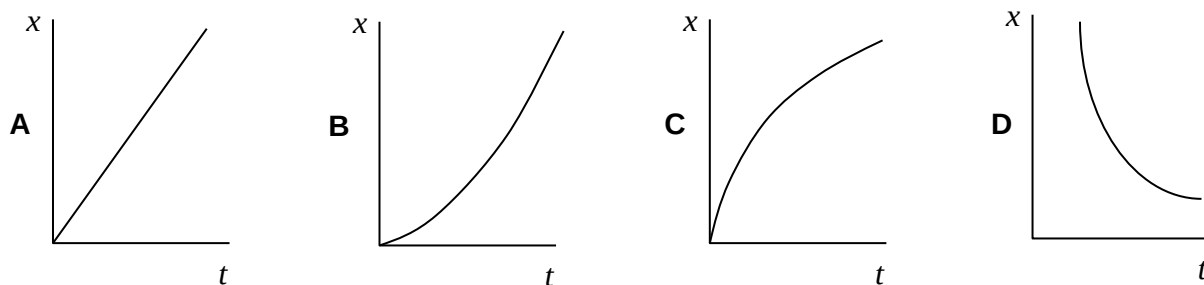
[1 mark]

- A** N m s^{-1} ☐
- B** $\text{kg m}^2 \text{s}^{-3}$ ☐
- C** J s^{-1} ☐
- D** $\text{kg m}^{-1} \text{s}^{-1}$ ☐

1	9
---	---

A car accelerates uniformly from rest along a straight road. Which graph shows the variation of displacement x of the car with time t ?

[1 mark]



- A** ☐
- B** ☐
- C** ☐
- D** ☐

2 0

Which of the following statements is correct?

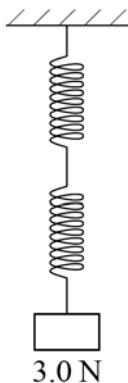
[1 mark]

The force acting on an object is equivalent to:

- A** its change of momentum. ☐
- B** the impulse it receives per second. ☐
- C** the energy it gains per second. ☐
- D** its acceleration per metre. ☐

2 1

A load of 3.0 N is attached to a spring of negligible mass and spring constant 15 N m^{-1} .



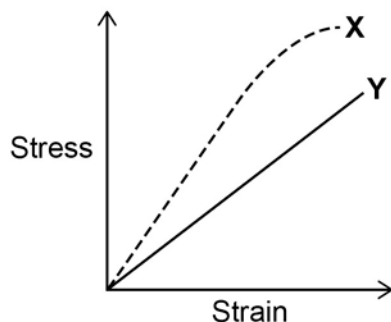
What is the energy stored in the spring?

[1 mark]

- A** 0.3 J ☐
- B** 0.6 J ☐
- C** 0.9 J ☐
- D** 1.2 J ☐

2 2

The diagram shows how the stress varies with strain for metal specimens X and Y which are different. Both specimens were stretched until they broke.



Which of the following is **not** correct?

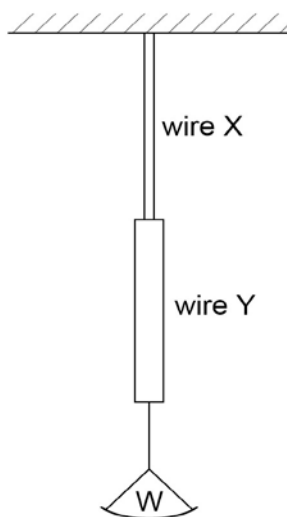
[1 mark]

- A X is stiffer than Y
- B X has a higher value of the Young modulus
- C X is more brittle than Y
- D Y has a lower maximum tensile stress than X

☐
☐
☐
☐

2 3

Two vertical copper wires X and Y of equal length are joined as shown. Y has a greater diameter than X. A weight W is hung from the lower end of Y.



Which of the following is correct?

[1 mark]

- A The strain in X is the same as that in Y
- B The stress in Y is greater than that in X
- C The tension in Y is the same as that in X
- D The elastic energy stored in X is less than that stored in Y

☐
☐
☐
☐

2	4
---	---

What is the relationship between the distance y travelled by an object falling freely from rest and the time x the object has been falling?

[1 mark]

A y is proportional to x^2

☐

B y is proportional to $\sqrt{x}$

☐

C y is proportional to $\frac{1}{x}$

☐

D y is proportional to $\frac{1}{x^2}$

☐

2	5
---	---

When an unstable nucleus, A_ZX , decays by emitting a β^- particle

CHECK QUESTION

[1 mark]

A A increases by 1, Z remains the same

☐

B A decreases by 1, Z remains the same

☐

C A remains the same, Z increases by 1

☐

D A remains the same, Z decreases by 1

☐